Outline Specifications — Cottages at Arcado Springs

Generated: 2026-08-28 Project Type: 55+ (HOPA) detached single-family cottage-home community, fee-simple lots with HOA-maintained yards — R-2 rezoning submission, City of Lilburn, Gwinnett County, Georgia (4535–4541 Arcado Rd SW, Land Lot 123, 6th District; 9.44 ac deeded) Divisions: 11 | Sections: 30 Status: **DRAFT — outline level, prepared by the Owner for the rezoning package** (Lilburn Zoning Ordinance 2023-603 §1003: "architectural renderings/elevations … colors and materials"). To be superseded by the Architect of Record's sealed construction documents and full-length specifications. Not for bidding or permit.

Basis of design (from `FACTS.md` §4, decided 2026-08-28)

- **Plans:** Plan A "The Springbrook" 38'-0" × 38'-0" body, 2 BR / 2 BA, ~1,444 SF conditioned; Plan B "The Laurel" 40'-0" × 40'-0" body, 2 BR + den / 2 BA, ~1,600 SF conditioned. One story, 9'-0" plate, ridge ≤ 24'-0"; 8:12 main gable, 4:12 porch; 6'-0"-deep covered front porch; rear porch/patio; 2-car garage (≥ 20'-0" × 21'-0") recessed ≥ 5'-0" behind the front wall (Table 4.2).
- **Lots:** typical 50'-0" × 100'-0" (5,000 SF); rear 20'-0" of perimeter lots held as undisturbed buffer easement (§313(1)). Private lane: 22'-0" pavement in a 31'-0" tract with 5'-0" sidewalk; hammerhead turnarounds; ~44 lots.
- **Universal design (voluntary — exceeds §734, which applies to attached dwellings only):** zero-step entries at porch and garage (finished floor within 1/2 in of the approach), 36-in doors, 60-in turning circles in baths, curbless showers, grab-bar blocking at every fixture, lever hardware, no interior thresholds, LED 3000K lighting, comfort-height toilets, hand-held showers. NFPA 13D sprinklers in every home (Division 21 — not in this outline).
- **Exterior character:** fiber-cement lap (6-in exposure) or board-and-batten; brick or stone water table (30 in) and porch piers; 8×8 cedar posts and brackets; architectural shingles. Three color schemes: (1) Warm White board-and-batten + stone; (2) Sage lap + brick; (3) Clay/Beige lap + stone.
- **Site elements:** 2 pickleball courts (each 20'-0" × 44'-0" court on a 30'-0" × 60'-0" fenced pad), village green, mail kiosk, monument entry sign ≤ 32 SF, 10-ft landscape strip along Arcado Rd, 20-ft buffer supplement, sod lawns.

Governing codes and standards (verified against Georgia DCA "Current State Minimum Codes for Construction", 2026-08-28)

- **2024 International Residential Code (IRC) with Georgia Amendments (2026)** — mandatory statewide 1 January 2026 (DCA memo, 9 Dec 2025). **The "IRC 2018 w/ GA amendments" assumed in the task brief and in earlier project notes is superseded**; IRC section numbers cited below follow the 2018–2024 numbering, which is unchanged for the provisions cited, but each shall be re-checked against the 2024 text and the 2026 Georgia amendments. Per the DCA Board's long-standing omission, the IRC's plumbing, electrical, and energy chapters are NOT used: the **2024 International Plumbing Code (GA Amendments 2026)**, **2023 National Electrical Code (GA Amendments 2026)**, and the IECC below govern instead.
- **2015 International Energy Conservation Code with Georgia Supplements and Amendments (2020, 2022, 2023)** — still the Georgia State Minimum Standard Energy Code; Georgia has not adopted the 2021 or 2024 IECC. Gwinnett County is **Climate Zone 3** (IECC Table R301.1). Georgia-amended Table R402.1.2, Zone 3: fenestration U-0.35 / SHGC-0.27; skylight U-0.55; **ceiling R-38** (R-30 accepted where full-height uncompressed R-30 extends over the top plate at the eaves, GA §R402.2.1); **wood-frame wall R-13** (Georgia amended Zones 2–4 to R-13 — the unamended 2015 IECC "R-20 or R-13 + R-5 ci" in the task brief does not apply); floor R-19; slab R-0; crawl 5/13; air leakage ≤ 5 ACH50 tested (GA §R402.4.1.2); ducts ≤ 4 cfm25/100 SF post-construction. The 2×6 R-20 wall specified here exceeds the Georgia minimum.
- **2024 International Fire Code** as adopted by the Georgia Safety Fire Commissioner (`FACTS.md` cites "IFC 2018 App. D" — update the citation); Gwinnett County Fire is the AHJ.
- **Accessibility:** the Georgia Accessibility Code (Ga. Comp. R. & Regs. 120-3-20) and the Fair Housing Act design standards do not apply to detached single-family dwellings; **ICC A117.1-2017** is used throughout as a voluntary design guide. HOPA 55+ per 42 U.S.C. §3607(b)(2)(C).
- **Local:** Lilburn Zoning Ordinance 2023-603 (Tables 4.1, 4.2; §313; §319; §1003); Lilburn Development Regulations (Code Appendix B — **not retrievable online; every citation marked VERIFY at the pre-application conference**); Lilburn Site Development Plan Review Checklist (DocumentCenter/1888); Gwinnett County UDO §900-70 (public-street pavement standard, used as the presumptive basis for the private lane); GDOT Standard Specifications, 2021 edition; Gwinnett County Stormwater Management Manual (2017).
- **Climatic and geographic design criteria (IRC Table R301.2, Gwinnett/Lilburn — VERIFY the locally adopted table):** ultimate design wind speed ≈ 115 mph, Exposure B; ground snow ≈ 5 psf; Seismic Design Category B; weathering "moderate"; frost line 12 in; termite "very heavy"; ice-barrier underlayment not required; FEMA Zone X (no SFHA on site).

- **Soils (NRCS via Gwinnett GIS):** Appling / Madison / Pacolet / Cecil-type Piedmont residual clays with a probable hydric unit near the stream head. A geotechnical report is required before foundation design (Division 31 — not in this outline).

Material mapping (pre-confirmed by the Owner, 2026-08-28)

Div	Section	Items
03	03 30 00 Cast-in-Place Concrete	monolithic slab-on-grade, turned-down footings, porch/garage slabs, zero-step approaches
04	04 20 00 Unit Masonry; 04 40 00 Stone Assemblies	brick veneer water table (scheme 2); stacked/ledge stone piers and water table (schemes 1, 3)
06	06 10 00 Rough Carpentry; 06 20 00 Finish Carpentry	2×6 walls, trusses, sheathing, blocking; cedar posts/brackets, painted trim
07	07 21 00; 07 27 00; 07 31 00; 07 46 46; 07 62 00; 07 71 00; 07 92 00	insulation; WRB/air barrier; shingles; fiber-cement siding; flashing; gutters; sealants
08	08 14 00 (with 08 16 13); 08 36 13; 08 53 13	entry and interior doors; carriage-style sectional garage door; vinyl (fiberglass alternate) windows
09	09 29 00; 09 30 13; 09 65 00; 09 91 13; 09 91 23	gypsum board; tile; LVP; exterior painting (three schemes); interior painting
10	10 14 00 Signage; 10 28 13 Toilet Accessories	monument entry sign (assigned here rather than 13 34 00 — it is a signage assembly, not a fabricated engineered structure); grab bars and accessories
12	12 36 00 Countertops	quartz / solid surface
22	22 40 00 Plumbing Fixtures	fixtures and trim only
26	26 50 00 Lighting	interior LED, porch and lane lighting (fixtures only)
32	32 12 16; 32 16 13; 32 31 13; 32 92 00; 32 93 00	lane paving; curb and sidewalks; pickleball fencing; sod; buffer and landscape plantings

General notes applying to every section: (0) Out of scope of this outline (listed for the roadmap): 21 13 00 NFPA 13D sprinklers; 23 00 00 HVAC; 26 00 00 wiring; 31 10 00 site clearing / 31 23 00 earthwork / 31 31 16 termite control; 32 18 23 pickleball court surfacing; 32 84 00 irrigation; 33 40 00 storm drainage and ponds. (a) Substitution requests shall be submitted in writing to the Architect a minimum of 10 days prior to bid date, with product data, samples, and a point-by-point comparison with the specified product. (b) "Appears consistent with" is used deliberately; nothing in this outline asserts code compliance. (c) Manufacturer lists are illustrative national/regional producers; Georgia-based producers are noted where they exist.

Division 03 — Concrete

Section 03 30 00 — Cast-in-Place Concrete

Part 1 — General

- **1.01 Section Includes:** Monolithic (turned-down) slab-on-grade foundations for each cottage; thickened slab edges and interior thickened slabs at bearing lines; garage slabs; porch and patio slabs; zero-step approach landings; brick/stone ledges; sign and pier footings.
- **1.02 Related Sections:** 04 20 00 / 04 40 00 (veneer ledges); 06 10 00 (anchor bolts, sill plates); 07 92 00 (isolation-joint sealant); 09 30 13 (recessed slab at curbless showers); 09 65 00 (slab moisture and flatness); 32 16 13 (site flatwork); 31 31 16 (termite pretreatment) and 31 23 00 (subgrade) — not in this outline.
- **1.03 References:** 2024 IRC (GA 2026) R401–R403.1 (footings), R403.1.4 (12-in minimum depth), R403.1.6 (anchorage), R404.1.6 / R317.1 (wood-to-grade clearances), R506 (slab-on-grade floors), R318 (termite protection); ACI 318-19 and ACI 332-20 (residential concrete); ASTM C94/C94M (ready-mixed concrete); ASTM C150/C150M (portland cement); ASTM C33 (aggregates); ASTM A615 (reinforcing bars); ASTM A1064 (welded wire reinforcement); ASTM E1745 (vapor retarders); ACI 302.1R (floor construction).
- **1.04 Submittals:** Mix designs; reinforcing shop drawings; vapor-retarder and admixture data; foundation plan sealed by a Georgia PE (DRAFT plans in this package are unsealed).
- **1.05 Quality Assurance:** Installer with 5 years' residential slab experience; geotechnical engineer to observe bearing at each footing excavation; slab elevations at every entry landing verified by survey before placement (zero-step tolerance ± 1/4 in).
- **1.06 Delivery, Storage, and Handling:** Reinforcing stored off the ground; concrete placed within 90 minutes of batching; no water added on site beyond the approved mix.
- **1.07 Warranty:** Contractor's 2-year workmanship warranty; structural warranty per the Georgia new-home warranty offered by the builder (10-year structural typical).

Part 2 — Products

- **2.01 Manufacturers (ready-mix):** Thomas Concrete (Atlanta); Argos USA; Heidelberg Materials; or approved equal.
- **2.02 Materials:** Normal-weight concrete, 3,000 psi minimum at 28 days per IRC Table R402.2 (weathering "moderate"); **3,500 psi, air-entrained 5–7 percent, at garage slabs, porches, patios, and exterior landings**; maximum 4-in slump (before plasticizer); footings 2-#4 continuous; slab 4 in minimum (R506.1) with 6×6-W1.4×W1.4 WWR at mid-depth or synthetic macro-fiber per Engineer; 10-mil Class A vapor retarder (ASTM E1745) directly under the slab (R506.2.3) over 4 in of clean crushed stone (R506.2.2); 1/2-in anchor bolts at 6 ft o.c. maximum and within 12 in of plate ends (R403.1.6).
- **2.03 Performance Criteria:** Footing bearing 12 in minimum below finished grade (frost/R403.1.4) and on undisturbed residual soil or engineered fill with allowable bearing per the geotechnical report (assume 2,000 psf until confirmed); slab flatness FF 25 / FL 20 minimum, with 3/16 in in 10 ft local tolerance at LVP areas (09 65 00); interior slab surface **within 1/2 in of the porch landing and garage approach at every entry (zero-step)**; exterior landings and garage aprons slope 1/8–1/4 in per ft (max 2 percent) away from the door for 3 ft minimum.
- **2.04 Finishes:** Interior slab steel-trowel; garage and exterior slabs medium broom; exposed edges 3/4-in chamfer; porch slab set 1/2 in below interior finished floor with beveled 1:2 transition at the door (ICC A117.1 §303).
- **2.05 Accessories:** Brick/stone ledge 4-1/2 in wide × 8 in deep formed into the turned-down edge at veneered walls; recessed shower area (slab dropped 1-1/2 in) at each bath; 1/2-in premolded isolation joint (ASTM D1751) at garage/house slab and at all flatwork abutting the foundation; sawcut control joints at ≤ 12 ft each way; termiticide pretreatment (R318) by the 31 31 16 applicator before the vapor retarder.

Part 3 — Execution

- **3.01 Examination:** Verify subgrade compaction (95 percent standard Proctor, ASTM D698, top 12 in) and geotechnical sign-off; verify plumbing/sprinkler sleeves, shower recess forms, and landing elevations against the approved zero-step entry detail.
- **3.02 Preparation:** Fine-grade stone base; lap vapor retarder 6 in and tape; set forms to ± 1/4 in; place and chair reinforcement; wet the subgrade in hot weather (ACI 305R).
- **3.03 Installation:** Place monolithically where practical; consolidate with internal vibration at turned-down edges; finish and cure 7 days (ASTM C309 curing compound compatible with LVP adhesives — or water cure); hold anchor bolts by template, not "wet set"; do not disturb termite-treated soil.
- **3.04 Field Quality Control:** Slump, air, and cylinder tests per ASTM C31/C39 for each 50 cy or each day's pour; survey check of finished-floor elevation at every entry before framing.
- **3.05 Cleaning and Protection:** Protect slab edges from backfill damage; keep slabs free of hydraulic-fluid and gypsum-mud staining; no vehicular loads on garage slabs for 7 days.

__[REVIEW REQUIRED]__ Structural values (bearing, reinforcement, slab thickness at thickened lines) are assumed pending the geotechnical report and a Georgia PE's foundation design. The zero-step detail must reconcile IRC R404.1.6 / R317.1 wood clearances and siding manufacturers' 2-in clearance to hardscape with the 1/2-in maximum rise at the door — see 04 40 00 and 07 46 46 for the masonry-base solution.

Division 04 — Masonry

Section 04 20 00 — Unit Masonry (Brick Veneer Water Table — Scheme 2)

Part 1 — General

- **1.01 Section Includes:** Anchored brick veneer water table, 30 in high above finished floor, on scheme 2 ("Sage") cottages, including porch pier bases where brick is used in lieu of stone; rowlock sill cap; brick at the scheme 2 monument sign.
- **1.02 Related Sections:** 03 30 00 (brick ledge); 06 10 00 (sheathing and tie backing); 07 27 00 (WRB behind veneer); 07 62 00 (through-wall flashing); 07 46 46 (siding above the cap); 07 92 00 (sealant at cap-to-siding joint); 10 14 00 (sign base).
- **1.03 References:** 2024 IRC (GA 2026) R703.8 (anchored masonry veneer: support, ties, air space, flashing, weeps); TMS 402/602-22 (Building Code Requirements and Specification for Masonry Structures); ASTM C216 (facing brick); ASTM C270 (mortar); ASTM C1714 (preblended mortar); ASTM A153 (galvanized ties); ASTM C1405 (glazed brick — not used); BIA Technical Notes 28B and 7.
- **1.04 Submittals:** Brick samples (3 units per range) and full-size mortar color samples; tie and flashing data; mock-up approval before production.
- **1.05 Quality Assurance:** Mason with 5 years' veneer experience; mock-up 4 ft × 3 ft including the ledge, flashing, weeps, rowlock cap, and the siding/sealant transition; retain as the standard for the work.

- **1.06 Delivery, Storage, and Handling:** Brick on pallets, covered, off the ground; mortar materials dry; no frozen or wet units laid.
- **1.07 Warranty:** Manufacturer's 50-year (or lifetime) limited brick warranty; Contractor's 2-year workmanship warranty.

Part 2 — Products

- **2.01 Manufacturers:** Cherokee Brick (Macon, GA); General Shale (Meridian Brick); Glen-Gery; Pine Hall Brick; or approved equal.
- **2.02 Materials:** Modular face brick (3-5/8 × 2-1/4 × 7-5/8 in), ASTM C216 Grade SW, Type FBS, red-brown blend as selected per scheme 2; mortar ASTM C270 Type N by proportion, colored to match; 22-ga × 7/8-in corrugated ties (or adjustable two-piece) hot-dip galvanized ASTM A153 Class B-2 at ≤ 32 in horizontal / ≤ 24 in vertical, ≤ 2.67 SF per tie (R703.8.4).
- **2.03 Performance Criteria:** 1-in nominal air space (R703.8.4); veneer bears fully on the concrete ledge (projection ≤ 1/3 of the unit thickness per TMS 402; support per R703.8.2); through-wall flashing and weeps at ≤ 33 in o.c. at the base (R703.8.6); maximum height as a water table 30 in, so no lintels or expansion joints are expected beyond corner control.
- **2.04 Finishes:** Rowlock (or cast-stone) sill cap sloped 15 degrees with drip; tooled concave joints; scheme 2 field brick with matching or lighter mortar as selected by the Architect from the manufacturer's full range.
- **2.05 Accessories:** Flexible flashing ASTM D1970 or 40-mil self-adhered with stainless drip edge; cotton-rope or cell weep vents; mortar-net collection device; rowlock cap flashing turned up 8 in behind the siding WRB.

Part 3 — Execution

- **3.01 Examination:** Verify ledge elevation, WRB coverage, and flashing sequence with 07 27 00 before laying; verify porch-pier footings.
- **3.02 Preparation:** Install through-wall flashing lapped over the WRB and extended 1/2 in past the face; set weeps; establish coursing so the cap lands at 30 in above finished floor.
- **3.03 Installation:** Lay in running bond, full head and bed joints, per TMS 602 Article 3; keep the air space clean; tie to studs through sheathing; tool joints when thumbprint hard; install the rowlock cap with full mortar joints and a sealant joint (07 92 00) at the siding above.
- **3.04 Field Quality Control:** Verify tie spacing and embedment (1-1/2 in minimum) before enclosure; check weep continuity with water test at the mock-up.
- **3.05 Cleaning and Protection:** Clean with bucket-and-brush method and proprietary cleaner per BIA Technical Note 20; no muriatic acid on colored mortar; cover walls at end of day.

__[REVIEW REQUIRED]__ Brick and mortar selections and the cap profile are generic pending scheme 2 samples; the veneer ledge and tie schedule shall be confirmed by the Engineer.

Section 04 40 00 — Stone Assemblies (Ledge/Stacked Stone Piers and Water Table — Schemes 1 and 3)

Part 1 — General

- **1.01 Section Includes:** Adhered ledge/stacked stone veneer (manufactured stone as the basis of design; quarried thin-veneer natural stone as an Architect-approved alternate) at the 30-in water table and at porch piers (nominal 20 × 20 in or 24 × 24 in) on schemes 1 ("Warm White") and 3 ("Clay/Beige"); stone-clad monument sign base for schemes 1 and 3; cast-stone pier caps and water-table sills.
- **1.02 Related Sections:** 03 30 00 (ledge, pier footings); 06 10 00 (pier framing or CMU cores); 06 20 00 (cedar posts bearing on pier caps); 07 27 00 (two WRB layers); 07 62 00 (flashing); 07 92 00 (sealants); 10 14 00 (sign).
- **1.03 References:** 2024 IRC (GA 2026) R703.12 (adhered masonry veneer installation and clearances) and R703.7 (lath/scratch coat by reference); TMS 402/602-22 Chapter 12 (adhered veneer); ASTM C1670 (adhered manufactured stone masonry veneer units); ASTM C1780 (installation practice for adhered manufactured stone veneer); ASTM C1063 (lath) and ASTM C847 (metal lath); ASTM C270 / C1714 (mortar); ASTM C1364 (architectural cast stone); MVMA Installation Guide (5th ed.); natural stone alternate: ASTM C1527 (adhered natural thin stone veneer).
- **1.04 Submittals:** 12 × 12 in stone boards (2 per scheme), cap samples, mortar color, lath/WRB data, ICC-ES report for manufactured stone; mock-up.
- **1.05 Quality Assurance:** Installer trained by the manufacturer; mock-up of one full pier and 4 ft of water table with cap, weep screed, and siding transition.
- **1.06 Delivery, Storage, and Handling:** Boxed units kept dry; blend from at least three boxes; no installation below 40 degrees F without protection.

- **1.07 Warranty:** Manufacturer's 50-year limited warranty (manufactured stone) or Contractor's 2-year workmanship warranty (natural stone).

Part 2 — Products

- **2.01 Manufacturers:** Eldorado Stone and Cultured Stone (Westlake Royal Building Products); ProVia (Hearth & Home); Coronado Stone Products; or approved equal. Natural thin-veneer alternate: Tennessee or North Georgia fieldstone ledge from an Architect-approved quarry.
- **2.02 Materials:** Ledge/stacked ("dry-stack" appearance, tight joint) profile, ≤ 15 psf installed (R703.12); scratch/setting mortar ASTM C270 Type S polymer-modified (ANSI A118.4-type "stone veneer mortar" acceptable); 2.5-lb self-furring galvanized lath (ASTM C847); two layers WRB per R703.12 (the outer layer may be the 07 27 00 housewrap; inner layer Grade D paper or a second housewrap); cast-stone caps ASTM C1364, 2-1/2 in thick, sloped with drip, buff or gray per scheme.
- **2.03 Performance Criteria:** Clearances per R703.12.1: 4 in above earth, 2 in above paved areas, **1/2 in above exterior walking surfaces supported by the same foundation** — which is the mechanism that lets the stone base meet the zero-step landing; weep screed at the base (ASTM C1063); stone freeze-thaw rated (ASTM C1670 §6).
- **2.04 Finishes:** Scheme 1: gray-tan "Appalachian"-type ledge blend; scheme 3: buff/tan ledge blend; caps as selected; joints raked/dry-stack; no sealer unless the manufacturer requires it.
- **2.05 Accessories:** Corner units; galvanized casing bead and weep screed; kick-out and cap flashings (07 62 00); 1/2-in setting-bed drainage mat where the manufacturer requires a rainscreen (recommended in Georgia's humid climate).

Part 3 — Execution

- **3.01 Examination:** Verify sheathing, WRB, flashing laps, and pier footings; verify landing elevations for the 1/2-in clearance.
- **3.02 Preparation:** Install second WRB layer, weep screed, lath (6-in laps, fasteners into studs at 6 in o.c.), and 1/2-in scratch coat; cure 48 hours.
- **3.03 Installation:** Per ASTM C1780 and the manufacturer's instructions: back-butter each unit, press with a slight twist, achieve 100 percent mortar contact; set caps in full bed with sealant at cap joints; keep stone 1/2 in clear of the landing and 2 in clear of walks.
- **3.04 Field Quality Control:** Random unit pull test (hand) at the mock-up; verify weep screed is open and unclogged after installation.
- **3.05 Cleaning and Protection:** Clean with soft brush and clean water while mortar is fresh; no acid; protect caps until Substantial Completion.

  [REVIEW REQUIRED] *Manufactured-vs-natural stone and cap profile are Owner/Architect selections pending samples; the pier core (CMU vs. framed) and post connection shall be detailed by the Engineer.*

Division 06 — Wood, Plastics, and Composites

Section 06 10 00 — Rough Carpentry (including 06 17 53 Shop-Fabricated Wood Trusses)

Part 1 — General

- **1.01 Section Includes:** 2×6 exterior wall framing at 16 in o.c., 9'-0" plate; 2×4 interior partitions; pressure-treated sill plates; wall and roof sheathing; pre-engineered metal-plate-connected roof trusses (8:12 main gables, 4:12 porch, raised heels); porch beams and posts backing; garage/house separation framing; **grab-bar and accessory blocking**; fire blocking and draft stopping; rough hardware.
- **1.02 Related Sections:** 03 30 00 (anchorage); 06 20 00 (cedar posts and trim); 07 21 00 (cavity insulation, raised-heel truss); 07 27 00 (WRB/air barrier continuity at plates); 07 31 00 (roof sheathing); 08 series (rough openings); 09 29 00 (garage separation board); 10 28 13 (grab-bar locations).
- **1.03 References:** 2024 IRC (GA 2026) R301.2 (design criteria), R302.6 and R302.11 (garage separation, fireblocking), R502–R602 (floor/wall framing), R602.10 (wall bracing), R802 (roof framing, uplift), R806 (attic ventilation), Table R602.3(1) (fastening); AWC NDS-2024 and WFCM-2024 (as referenced by the 2024 IRC — VERIFY edition); ANSI/TPI 1-2022 (metal-plate-connected wood trusses); APA PRP-108 / DOC PS 2 (sheathing); AWPA U1 (preservative treatment, UC2 for sills on concrete); SPIB / WWPA grading rules; ICC A117.1-2017 §604.5, §608.3, §609 (grab-bar geometry for blocking layout).
- **1.04 Submittals:** Truss layout and truss design drawings sealed by a Georgia PE with reactions, uplift, and bracing; sheathing/nailing plan for wall bracing; blocking plan for each bath (grab bars, seats, slide bars, towel bars); treated-lumber certifications.

- **1.05 Quality Assurance:** Framer with 5 years' single-family experience; truss fabricator certified under the SBCA in-plant QC program; pre-drywall inspection of blocking, fireblocking, and air-sealing before insulation.
- **1.06 Delivery, Storage, and Handling:** Lumber stacked and covered; trusses stored flat and braced; treated lumber separated from untreated.
- **1.07 Warranty:** Contractor's 2-year workmanship; builder's 10-year structural warranty; truss manufacturer's design warranty.

Part 2 — Products

- **2.01 Manufacturers:** Trusses: Builders FirstSource, UFP Industries, 84 Lumber components (regional plants); engineered lumber: Weyerhaeuser Trus Joist, Boise Cascade, LP Building Solutions; sheathing: Georgia-Pacific (GA), Huber (ZIP System alternate with 07 27 00), LP; connectors: Simpson Strong-Tie, MiTek; or approved equal.
- **2.02 Materials:** Studs SPF or SYP No. 2, 2×6 at 16 in o.c. (104-5/8-in precut for 9'-0" plate); double top plates; SYP No. 2 headers or LVL; PT sill plate (AWPA UC2 minimum, UC4A at grade-level exposure) with foam sill sealer; wall sheathing 7/16-in OSB (or 15/32-in plywood) APA-rated, Exposure 1, continuous (CS-WSP bracing, R602.10.4); roof sheathing 15/32-in APA-rated with H-clips at 24 in o.c. trusses; trusses SYP with 20-ga plates per TPI 1; blocking 2×10 flat (or 3/4-in plywood 12 in wide) let in flush.
- **2.03 Performance Criteria:** Loads per IRC R301.2 and the Engineer: roof live 20 psf, ground snow 5 psf, wind 115 mph ultimate Exposure B (VERIFY), SDC B; uplift connectors at every truss bearing per truss reactions (R802.11); porch trusses/beams sized for the 6-ft porch and the 8×8 posts; attic ventilation 1/300 with balanced soffit/ridge (R806.2); raised-heel trusses ≥ 12 in at exterior walls to carry full-depth R-38 over the plate (07 21 00); garage ceiling trusses designed for the 1-hr-equivalent gypsum weight (not rated — R302.6 separation only).
- **2.04 Finishes:** Not applicable (concealed); exposed porch framing wrapped per 06 20 00.
- **2.05 Accessories:** Hot-dip galvanized fasteners at treated lumber and exterior (ASTM A153); hurricane ties; anchor-bolt plate washers 3 × 3 × 0.229 in; fireblocking mineral wool or wood per R302.11; foam gaskets at plates for air sealing (IECC Table R402.4.1.1).

Part 3 — Execution

- **3.01 Examination:** Verify foundation dimensions, anchor bolt positions, and slab elevations at every door (zero-step); verify truss bearing walls match the truss layout.
- **3.02 Preparation:** Snap layout; set sills level on sealer; shim ≤ 1/4 in; lay out bath blocking from the grab-bar plan: **continuous horizontal blocking centered 33–36 in above finished floor on all shower/tub and toilet walls, vertical blocking for 18-in vertical bars, and blocking at 30–48 in for slide bars, fold-down seats (rated 250 lb, A117.1 §609.8), and towel bars**.
- **3.03 Installation:** Frame per IRC Chapter 6 and Table R602.3(1); sheathe walls continuously with edge nailing 6 in / field 12 in (or per bracing plan); set trusses per the sealed layout with temporary and permanent bracing per BCSI; install uplift connectors before roof sheathing; frame garage/house wall and ceiling continuous for the 1/2-in gypsum separation (R302.6); fireblock at plates, soffits, and chases; leave 1/2-in gaps at slab edges for foam sealing.
- **3.04 Field Quality Control:** Third-party or AHJ framing inspection; truss bracing inspection by the fabricator's engineer where required; blocking photographed and logged per lot before drywall (HOA maintenance record for future grab-bar installation).
- **3.05 Cleaning and Protection:** Keep sheathing dry (cover roof within 7 days); remove debris from stud cavities before insulation; protect PT cut ends with field treatment (AWPA M4).

__[REVIEW REQUIRED]__ All structural loads, bracing, and connections are presumptive until designed by the Georgia PE; confirm the 2024 IRC/2026 GA amendment section numbers and the NDS/WFCM editions referenced.

Section 06 20 00 — Finish Carpentry

Part 1 — General

- **1.01 Section Includes:** 8×8 nominal Western Red Cedar porch posts on stone/brick piers; cedar gable brackets/corbels; cedar-wrapped porch beams; painted exterior trim (cellular PVC or fiber-cement per 07 46 46); porch ceilings (beadboard); interior casing, base, window stools, closet shelving; door and cabinet blocking.
- **1.02 Related Sections:** 04 40 00 (pier caps); 06 10 00 (structural posts/beams); 07 46 46 (fiber-cement trim); 07 62 00 (post-base and beam flashing); 07 92 00; 09 91 13 / 09 91 23 (finishes).
- **1.03 References:** AWI/AWMAC/WI Architectural Woodwork Standards (AWS) 2nd ed., Custom grade; WRCLA (Western Red Cedar Lumber Association) grading and installation guides; WCLIB/WWPA grading rules; ASTM D4216 (rigid PVC compounds; cellular PVC trim per the manufacturer's ICC-ES report); 2024 IRC (GA 2026) R317.1 (protection of wood; post-to-pier separation), R407.3 (post-base connection); ANSI A208.2 (MDF).
- **1.04 Submittals:** Cedar grade certification and sample; bracket shop drawing (one per plan); PVC/fiber-cement trim profiles; interior trim profiles; post-base hardware data.

- **1.05 Quality Assurance:** Carpenter with 5 years' finish experience; first cottage porch serves as the mock-up for post, bracket, and beam wrap.
- **1.06 Delivery, Storage, and Handling:** Cedar stickered and covered, acclimated 7 days; interior trim delivered after the building is dried-in and conditioned.
- **1.07 Warranty:** Contractor's 2-year workmanship; cellular PVC manufacturer's 25-year (or lifetime) limited warranty.

Part 2 — Products

- **2.01 Manufacturers:** Cedar: WRCLA-member mills; cellular PVC trim: AZEK (Timbertech), Versatex, Kleer (Tapco); interior trim: Woodgrain, Metrie, Boise Cascade; or approved equal.
- **2.02 Materials:** Posts 8×8 nominal (7-1/2 in actual) Western Red Cedar, WRCLA "Appearance"/No. 1 grade, S4S or resawn, ends sealed; alternate: 6×6 PT structural post (06 10 00) wrapped in 1×10 clear cedar with mitered corners; brackets 4×6 and 4×8 cedar, glued and through-bolted; porch beam wrap 1× cedar or fiber-cement; exterior painted trim 5/4 × 4 and 5/4 × 6 cellular PVC (ASTM D4216 compound) or fiber-cement trim (07 46 46); porch ceiling 5/16-in fiber-cement beadboard or 1×6 T&G cedar; interior casing 3-1/2 in and base 5-1/4 in primed finger-jointed pine or MDF, shaker/flat profile; stools solid poplar.
- **2.03 Performance Criteria:** Posts separated from the masonry cap by a standoff base ≥ 1 in (R317.1.4) with a 1-in-diameter minimum uplift/lateral connection (R407.3) rated by the Engineer; brackets non-structural (decorative) unless noted.
- **2.04 Finishes:** Cedar: semi-transparent penetrating stain (09 91 13) "natural cedar" on schemes 1 and 3, painted trim color on scheme 2 as selected; painted trim per scheme (09 91 13); interior trim semi-gloss white (09 91 23).
- **2.05 Accessories:** Stainless or hot-dip galvanized post base/cap (Simpson ABU/CPS-type or approved equal); stainless finish nails at cedar; PVC cement and color-matched fasteners at PVC; construction adhesive.

Part 3 — Execution

- **3.01 Examination:** Verify pier cap elevations and beam heights; verify wall trim backing.
- **3.02 Preparation:** Back-prime all cedar and paint-grade trim; seal cut ends; lay out post centerlines from the porch beam.
- **3.03 Installation:** Set posts plumb on standoff bases; cut to bearing with full contact; install brackets with concealed bolts and caps; run exterior trim with scarfed joints over backing, 1/8-in sealant gap at siding; interior trim with tight miters, nails set and filled, caulked to wall.
- **3.04 Field Quality Control:** Post plumb ± 1/8 in over the height; reveal lines consistent ± 1/16 in across each porch.
- **3.05 Cleaning and Protection:** Protect cedar from mortar and paint splatter; protect interior trim until final painting.

  [REVIEW REQUIRED] *Post size versus structural demand (8×8 is architectural; a 6×6 PT wrapped post may be the economical structure) and bracket attachment shall be confirmed by the Engineer.*

Division 07 — Thermal and Moisture Protection

Section 07 21 00 — Thermal Insulation

Part 1 — General

- **1.01 Section Includes:** Wall cavity insulation (2×6, R-20); attic insulation (R-38 blown); insulation at the garage/house wall and any conditioned-space ceiling over the garage; eave baffles; insulation certificate.
- **1.02 Related Sections:** 06 10 00 (raised-heel trusses, baffles backing); 07 27 00 (air barrier — insulation performance depends on it); 09 29 00 (interior air barrier at ceilings).
- **1.03 References:** 2015 IECC with Georgia Amendments (2020–2023) Table R402.1.2 (Zone 3: wall R-13, ceiling R-38, floor R-19, slab R-0), §R402.2.1 (R-30 uncompressed alternative), §R401.3 (certificate), §R303.1 (installation and marking); ASTM C665 (mineral-fiber batts), ASTM C764 (loose-fill mineral fiber), ASTM C739 (cellulose), ASTM C1029 (spray polyurethane, if used); ASTM E84 (surface burning); RESNET/ANSI 301 Grade I installation.
- **1.04 Submittals:** Product data with R-value per inch, bag-count chart for blown attic insulation, and the IECC R401.3 certificate posted at the electrical panel.
- **1.05 Quality Assurance:** Installer certified by the manufacturer; Grade I installation verified by the HERS/energy rater at the pre-drywall inspection.
- **1.06 Delivery, Storage, and Handling:** Keep dry and in packaging; do not install wet batts.
- **1.07 Warranty:** Manufacturer's limited lifetime (batts); Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** Owens Corning; Johns Manville; Knauf Insulation; CertainTeed; cellulose alternate Greenfiber; or approved equal.
- **2.02 Materials:** Walls: R-20 unfaced fiberglass batts (ASTM C665 Type I) friction-fit in 2×6 cavities, or R-21 high-density, or blown-in-blanket fiberglass at R-23; attic: blown fiberglass (ASTM C764 Type I) or cellulose (ASTM C739) to R-38 minimum (R-49 optional upgrade); garage/house wall R-20; attic access hatch insulated R-38 with gasket; kraft-faced batts permitted only where a Class III vapor retarder is acceptable (Zone 3: no interior vapor retarder required, IRC R702.7).
- **2.03 Performance Criteria:** Whole-house target: exceed GA Table R402.1.2 by ≥ 50 percent at walls (R-20 vs R-13) and meet R-38 at 100 percent of the ceiling using raised-heel trusses; Grade I; ENERGY STAR Single-Family New Homes v3.2 optional (VERIFY Owner interest).
- **2.04 Finishes:** Not applicable.
- **2.05 Accessories:** Cardboard or plastic eave baffles at every truss bay with 2-in clear vent path; attic rulers (one per 300 SF); insulation dams at the hatch and at eaves; metal-faced fire-rated tape at spray foam if used.

Part 3 — Execution

- **3.01 Examination:** Verify air-sealing (07 27 00) is complete and inspected; verify baffles and dams; verify no wet framing (≤ 19 percent MC).
- **3.02 Preparation:** Install baffles and dams; mark attic depth on rulers; seal top plates and penetrations before blowing.
- **3.03 Installation:** Batts cut to fit around wiring/boxes without compression (Grade I); split around cables; blow attic to the settled depth on the bag chart; keep insulation 3 in from non-IC recessed fixtures (all fixtures IC-rated per 26 50 00, so not expected).
- **3.04 Field Quality Control:** Rater's Grade I inspection with photos; depth probe at 10 attic locations per house; certificate posted.
- **3.05 Cleaning and Protection:** Remove overspray/loose fill from soffit vents; protect batts from moisture until drywall.

  [REVIEW REQUIRED] *Georgia has not adopted the 2021/2024 IECC; if adoption occurs before permit, wall R-values and ACH50 limits change. Confirm the energy-code path (prescriptive vs. ERI) with the rater.*

Section 07 27 00 — Air Barriers (Weather-Resistive Barrier and Air Barrier)

Part 1 — General

- **1.01 Section Includes:** Mechanically attached drainable housewrap as the water-resistive barrier (WRB) and exterior air barrier; self-adhered flashing at openings; sill pans; sealing of the ceiling plane, plates, and penetrations to form a continuous air barrier; blower-door testing.
- **1.02 Related Sections:** 04 20 00 / 04 40 00 (WRB behind veneer, two layers at stone); 06 10 00 (sheathing); 07 46 46; 07 62 00; 07 92 00; 08 53 13 / 08 14 00 (flashed openings); 09 29 00 (airtight drywall).
- **1.03 References:** 2024 IRC (GA 2026) R703.2 (WRB) and R703.4 (flashing); 2015 IECC (GA) §R402.4.1.1 and Table R402.4.1.1 (air barrier and insulation installation), §R402.4.1.2 (≤ 5 ACH50, tested); ASTM E1677 (Type I WRB); ICC-ES AC38; ASTM E2273 (drainage efficiency); ASTM E2112 (window/door installation); AAMA 711 (self-adhered flashing), AAMA 714 (sill pans); ASTM E779 / RESNET (blower door).
- **1.04 Submittals:** WRB and flashing data with ICC-ES reports; sequencing details at window sill/head, stone base, kick-out, and porch roof-to-wall; blower-door test reports per house.
- **1.05 Quality Assurance:** Installer trained by the WRB manufacturer; the first cottage is a full-envelope mock-up reviewed before the WRB is covered; independent RESNET/BPI rater performs testing.
- **1.06 Delivery, Storage, and Handling:** Rolls stored out of sunlight; WRB exposure limited to the manufacturer's UV limit (typically 120 days) before siding.
- **1.07 Warranty:** Manufacturer's 10-year limited (WRB) system warranty where offered; Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** DuPont (Tyvek DrainWrap / HomeWrap); Kingspan (GreenGuard DrainWrap); Benjamin Obdyke (HydroGap); Henry/Carlisle (Blueskin VP100 self-adhered alternate); integrated-sheathing alternate: Huber ZIP System or Georgia-Pacific ForceField (coordinate with 06 10 00); or approved equal.
- **2.02 Materials:** Drainable housewrap, ASTM E1677 Type I, drainage efficiency ≥ 90 percent (ASTM E2273), UV-stable, perm ≥ 10; 4-in and 6-in self-adhered butyl or acrylic flashing (AAMA 711 Level 3); preformed or site-formed sill pans (AAMA 714); cap-nail fasteners; manufacturer's seam tape.
- **2.03 Performance Criteria:** Continuous air barrier at the exterior sheathing plane and the ceiling gypsum plane, tied at top plates; ≤ **5 ACH50** (Georgia) with a project target of ≤ 3 ACH50; water-shedding laps ≥ 6 in horizontal, ≥ 6 in vertical; every

opening flashed "shingle fashion" with sill pan, jamb, head, and the WRB lapped over the head flashing.

- **2.04 Finishes:** Not applicable (concealed).
- **2.05 Accessories:** Spray foam (low-expansion) and acoustical/air-seal sealant at plates, penetrations, and window shims; gaskets at electrical boxes; fire-rated sealant at chases (R302.11).

Part 3 — Execution

- **3.01 Examination:** Verify sheathing is fastened, dry, and free of gaps > 1/4 in; verify rough openings are square and sized for the sill pan.
- **3.02 Preparation:** Install sill pans first; cut the WRB with the modified "I" cut at openings; install stone-base weep screed and second WRB layer where 04 40 00 requires.
- **3.03 Installation:** Wrap from the bottom up, lapped shingle fashion, fastened at 12–18 in o.c. to studs with cap nails; tape all seams and penetrations; integrate with 07 62 00 kick-out and step flashing and with veneer through-wall flashing; seal top plates to sheathing and to ceiling gypsum (airtight-drywall approach) per Table R402.4.1.1; seal every wire, pipe, duct, and sprinkler penetration.
- **3.04 Field Quality Control:** Pre-siding inspection with photo log of every opening; blower-door test at completion (and optional mid-construction test at drywall) documented on the IECC certificate; failed houses re-sealed and re-tested.
- **3.05 Cleaning and Protection:** Repair tears with tape the same day; cover the WRB with siding within the exposure limit.

__[REVIEW REQUIRED]__ Generic system pending Architect's selection between housewrap and integrated-sheathing WRB; the stone-base and zero-step flashing details require the Architect's drawn detail.

Section 07 31 00 — Shingles and Shakes (Architectural Asphalt Shingles)

Part 1 — General

- **1.01 Section Includes:** Laminated ("architectural"/dimensional) asphalt shingles, Class A, on 8:12 main roofs and 4:12 porch roofs; underlayment; self-adhered membrane at eaves, valleys, and penetrations; starter and hip/ridge shingles; ridge and soffit ventilation products; roof accessories.
- **1.02 Related Sections:** 06 10 00 (sheathing, ventilation area); 07 62 00 (drip edge, step and kick-out flashing, valley metal); 07 71 00 (gutters); 07 92 00.
- **1.03 References:** 2024 IRC (GA 2026) R905.1–R905.2 (asphalt shingles: slope, underlayment, fastening, flashing), R903.2 (flashing), R806 (ventilation); ASTM D3462 (asphalt shingles, glass felt); ASTM E108 / UL 790 Class A; ASTM D7158 Class H (wind, 150 mph) and ASTM D3161 Class F; ASTM D8257 (mechanically attached polymeric underlayment) or ASTM D226 Type II felt; ASTM D1970 (self-adhered underlayment); ASTM D4869; ARMA Residential Asphalt Roofing Manual; NRCA Roofing Manual — Steep-slope.
- **1.04 Submittals:** Shingle samples (3 colors per scheme), underlayment data, ventilation calculation (net free area per R806), warranty sample, fastening pattern.
- **1.05 Quality Assurance:** Roofer certified by the manufacturer (for the system warranty); mock-up: first cottage roof reviewed for starter, valley, and ridge details.
- **1.06 Delivery, Storage, and Handling:** Bundles flat, covered, ≤ 2 pallets high; no installation of shingles below 40 degrees F without hand-sealing.
- **1.07 Warranty:** Manufacturer's limited lifetime (minimum 30-year) shingle warranty with **10-year non-prorated period, 130-mph wind coverage (with specified starter/ridge), and 10-year algae-resistance**; Contractor's 5-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** GAF (Timberline HDZ); Owens Corning (Duration); CertainTeed (Landmark); Atlas Roofing (Pinnacle Pristine, Atlanta GA); or approved equal.
- **2.02 Materials:** Laminated fiberglass-mat asphalt shingles, ASTM D3462, Class A, algae-resistant (copper-bearing granules), with manufacturer's starter strip and matching hip/ridge cap; underlayment: synthetic ASTM D8257 (one layer at 8:12; two layers or self-adhered at 4:12 per R905.2.7 — VERIFY), self-adhered ASTM D1970 36 in at eaves, valleys, and around penetrations (ice barrier not required in Zone 3 but specified for durability); ridge vent shingle-over, ~18 in² NFA/ft, with external baffle.
- **2.03 Performance Criteria:** Class A; wind ASTM D7158 Class H; nail pattern per manufacturer's high-wind (6-nail) pattern; 12-ga galvanized or stainless roofing nails, 3/8-in heads, penetrating ≥ 3/4 in or through sheathing (R905.2.5); balanced ventilation 1/300 with ≥ 40 percent at the ridge (R806.2).
- **2.04 Finishes:** Scheme 1 charcoal; scheme 2 weathered wood; scheme 3 driftwood/brown — as selected by the Architect from the manufacturer's full range; cool-roof color not required.

- **2.05 Accessories:** Painted-steel or aluminum drip edge (07 62 00); lead-free flexible pipe boots (EPDM/aluminum) rated 20 years; roof-to-wall kick-out diverters; sprinkler-system and bath-fan roof jacks with backdraft dampers; snow guards not required.

Part 3 — Execution

- **3.01 Examination:** Verify sheathing is dry, fastened, with H-clips and 1/8-in gaps; verify baffles, ridge slot (2 in), and drip edge sequence (under underlayment at eaves, over at rakes — R905.2.8.5).
- **3.02 Preparation:** Install self-adhered membrane at eaves/valleys; install underlayment with 4-in horizontal / 6-in end laps, cap-nailed; install step/kick-out flashing bases with 07 62 00.
- **3.03 Installation:** Per R905.2, ARMA, and the manufacturer: starter strip at eaves and rakes; 5-5/8-in exposure (or per product); closed-cut or woven valleys over metal; 6 nails per shingle; hand-seal at rakes and at 4:12 porches in cold weather; ridge vent continuous with end plugs; cap shingles fastened with 1-3/4-in nails.
- **3.04 Field Quality Control:** Manufacturer's inspection where required for the system warranty; visual check of nail placement (no exposed, high, or overdriven nails) on 10 percent of each roof.
- **3.05 Cleaning and Protection:** Remove loose granules and nails from gutters and grounds (magnetic sweep); no traffic on finished roofs above 80 degrees F.

[REVIEW REQUIRED] Life-safety adjacent (Class A rating, wind attachment); underlayment requirement at 4:12 and the locally adopted wind speed to be confirmed by the Architect against the 2024 IRC/GA amendments.

Section 07 46 46 — Fiber-Cement Siding

Part 1 — General

- **1.01 Section Includes:** Fiber-cement lap siding, 7-1/4-in plank at 6-in exposure (schemes 2 and 3); fiber-cement vertical panels with battens (board-and-batten, scheme 1); fiber-cement trim, corner boards, frieze, soffit, and porch-ceiling beadboard; factory or field finish.
- **1.02 Related Sections:** 04 20 00 / 04 40 00 (water-table cap transition); 06 10 00 (sheathing, stud backing at butt joints); 06 20 00 (cedar and PVC trim); 07 27 00 (WRB); 07 62 00 (Z- and head flashing); 07 92 00; 09 91 13 (paint).
- **1.03 References:** 2024 IRC (GA 2026) Table R703.3(1) (fiber-cement siding attachment) and R703.10 (fiber-cement siding), R703.4 (flashing); ASTM C1186 (Grade II, Type A fiber-cement flat sheets/siding); ASTM E84 (Class A); ASTM E136 (noncombustible); ICC-ES ESR-2290 (James Hardie) or equivalent evaluation report; manufacturer's installation requirements for the HZ10 (hot-humid) climate zone.
- **1.04 Submittals:** Product data, texture and profile samples (12 in), batten spacing elevation for scheme 1, ColorPlus or field-paint color chips, fastener schedule, flashing details at butt joints and the water-table cap.
- **1.05 Quality Assurance:** Installer with 3 years' fiber-cement experience and manufacturer training (James Hardie "Preferred" or equivalent); mock-up 8 ft × 8 ft per scheme with trim, corner, batten, and stone/brick base transition.
- **1.06 Delivery, Storage, and Handling:** Store flat, covered, off the ground; keep dry before installation (wet boards shrink after installation); carry planks on edge; use HEPA-vacuum dust-collecting saws (OSHA silica rule 29 CFR 1926.1153).
- **1.07 Warranty:** Manufacturer's 30-year non-prorated substrate warranty and 15-year finish warranty (factory finish); Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** James Hardie Building Products (HardiePlank, HardiePanel, HardieTrim — HZ10); Nichiha USA (Johns Creek, GA; Sierra Premium lap and panels); Allura (Elementia); or approved equal.
- **2.02 Materials:** ASTM C1186 Grade II Type A; lap: 5/16 × 7-1/4-in plank, smooth or "cedarmill" texture as selected, installed at 6-in exposure (1-1/4-in overlap); board-and-batten: 5/16 × 48 × 108/120-in smooth vertical panels with 5/4 × 2-1/2-in fiber-cement battens at 12 in o.c. (16 in o.c. at the Architect's option); trim 5/4 × 4 (corners, 3-1/2-in exposed face) and 5/4 × 6 (windows/doors) with 1-in drip cap; frieze 5/4 × 8; vented soffit 1/4-in panels; porch ceiling 1/4-in beadboard.
- **2.03 Performance Criteria:** Class A (E84), noncombustible (E136); wind attachment per ESR and Table R703.3(1) for 115 mph Exposure B (VERIFY): blind-nailed 6d (2-in) hot-dip galvanized or stainless siding nails into studs at 16 in o.c., 1 in from top edge; clearances: **6 in to finished grade, 2 in to roofs/paving/decks, 1/4 in to horizontal flashing, 1/8 in to trim (sealed)** — at zero-step landings the siding stops on the masonry water table/cap so the 2-in hardscape clearance is honored by the stone (04 40 00), not by the siding.
- **2.04 Finishes:** Factory ColorPlus (preferred for the HOA's maintenance cycle) or field-painted per 09 91 13: scheme 1 warm white B&B; scheme 2 sage lap; scheme 3 clay/beige lap; trim warm white on all schemes unless the Architect selects otherwise.

- **2.05 Accessories:** Pre-formed galvanized or color-matched joint flashing (6 in wide) behind every butt joint; Z-flashing at horizontal trim and the water-table cap; color-matched touch-up kits; corner posts; sealant (07 92 00, paintable, ASTM C920 Class 25).

Part 3 — Execution

- **3.01 Examination:** Verify WRB is complete, taped, and flashed; verify the water-table cap and its flashing are installed; verify stud layout for panel edges and butt joints.
- **3.02 Preparation:** Snap level lines at each course; install Z-flashing over the cap and over all horizontal trim; install starter strip for the first lap course; lay out battens symmetrical about openings.
- **3.03 Installation:** Per the ESR and manufacturer's instructions: butt joints in moderate contact over studs with joint flashing (no caulk at flashed joints); panels vertical with edges on studs, joints under battens; 1/8-in gaps at trim, sealed; no face-nailing except at the top course and where required; factory-primed cut edges field-primed; keep required clearances; install trim before siding (Hardie "trim-first") with siding butted to trim.
- **3.04 Field Quality Control:** Check exposure uniformity (± 1/16 in per course), nail depth (flush, not overdriven), and clearances on every elevation before painting.
- **3.05 Cleaning and Protection:** Wash dust from the surface before painting; touch up cut edges; protect from mortar and stain splatter at stone piers and cedar posts.

__[REVIEW REQUIRED]__ Fastener schedule and exposure/wind values depend on the adopted wind speed and product ESR; the batten spacing and trim widths are placeholders pending the Architect's elevations.

Section 07 62 00 — Sheet Metal Flashing and Trim

Part 1 — General

- **1.01 Section Includes:** Roof drip edge; step, apron, and kick-out flashing at porch and gable-return roof-to-wall conditions; valley metal; head/drip caps over windows, doors, and horizontal trim; Z-flashing; through-wall flashing drip at veneer; post-base and pier-cap flashings; garage-door head flashing.
- **1.02 Related Sections:** 04 20 00 / 04 40 00; 07 27 00 (self-adhered membrane flashing); 07 31 00; 07 46 46; 07 71 00; 07 92 00.
- **1.03 References:** 2024 IRC (GA 2026) R703.4 (wall flashing locations), R903.2 (roof flashing), R905.2.8 (asphalt shingle flashing: base, step, kick-out, drip edge); SMACNA Architectural Sheet Metal Manual, 8th ed.; ASTM A653/A653M (G90 galvanized steel); ASTM B209 (aluminum sheet); ASTM A240 (stainless); AAMA 2605 (fluoropolymer finish, where pre-finished).
- **1.04 Submittals:** Profiles and gauges; finish color chips; shop drawings for kick-out and pier-cap flashings.
- **1.05 Quality Assurance:** Installed by the roofer and the siding installer under one sequencing plan; reviewed at the 07 27 00 envelope mock-up.
- **1.06 Delivery, Storage, and Handling:** Keep pre-finished metal in protective film until installed; store dry to prevent wet-storage stain.
- **1.07 Warranty:** Finish warranty 20 years (AAMA 2605 fluoropolymer) or 10 years (polyester); Contractor's 5-year workmanship (with roofing).

Part 2 — Products

- **2.01 Manufacturers:** Berger Building Products; Amerimax Home Products; Petersen Aluminum (PAC-CLAD); Drexel Metals; or approved equal.
- **2.02 Materials:** Pre-finished aluminum 0.024 in (or 26-ga G90 galvanized steel, pre-finished) for exposed flashings; 0.032-in aluminum at kick-outs and pier caps; stainless steel Type 304, 26 ga, where in contact with treated lumber, mortar, or copper-bearing algae-resistant shingle runoff; galvanized steel step flashing 4 × 4 × 8 in minimum (R905.2.8.3); drip edge with 2-in roof leg and 1-1/2-in face.
- **2.03 Performance Criteria:** No dissimilar-metal contact (isolate aluminum from mortar, treated lumber, and steel fasteners); hemmed exposed edges; expansion joints at ≤ 10 ft runs of continuous trim; kick-out flashing ≥ 4 in high directing water into the gutter, never behind siding.
- **2.04 Finishes:** Color-matched to trim or roof per scheme; concealed flashings mill finish.
- **2.05 Accessories:** Compatible fasteners (stainless at aluminum), sealant per 07 92 00, butyl tape at laps.

Part 3 — Execution

- **3.01 Examination:** Verify WRB and underlayment sequence allows shingle-fashion laps; verify porch roof/wall intersections have backing for kick-outs.
- **3.02 Preparation:** Cut and form on site or shop; pre-bend kick-outs; isolate metals with tape or coating.
- **3.03 Installation:** Per SMACNA and the referenced IRC sections: drip edge under underlayment at eaves and over at rakes; step flashing one piece per course, counter-flashed by the siding with a 1-in gap to the roof; kick-out at every roof-to-wall eave termination; head flashing over every opening lapped by the WRB; Z-flashing over horizontal trim with end dams; pier caps sloped with drips over the stone.
- **3.04 Field Quality Control:** Water test the porch roof-to-wall and one kick-out at the mock-up house; inspect all head flashings before siding closes them.
- **3.05 Cleaning and Protection:** Remove strippable film; remove metal filings from roofs and gutters to prevent rust staining.

[REVIEW REQUIRED] Generic profiles; the Architect shall detail the kick-out, water-table cap, and pier-cap flashings for the drawings.

Section 07 71 00 — Roof Specialties (Gutters and Downspouts)

Part 1 — General

- **1.01 Section Includes:** Seamless aluminum K-style gutters, downspouts, hangers, outlets, and splash blocks or connections to lot drainage; gutter guards (optional HOA upgrade).
- **1.02 Related Sections:** 07 31 00 (drip edge into gutter); 07 62 00; 07 92 00; 32 16 13 (walk-side downspout discharge); 33 40 00 storm laterals (not in this outline — required where lots drain to the lane system).
- **1.03 References:** SMACNA Architectural Sheet Metal Manual, 8th ed.; ASTM B209 (aluminum alloy 3105-H24); ASTM B221 (extrusions); 2024 IRC (GA 2026) R801.3 (roof drainage discharged ≥ 5 ft from foundation or to an approved drainage system); Gwinnett County Stormwater Management Manual (lot drainage, 2017).
- **1.04 Submittals:** Gutter profile and gauge, color chip, downspout layout on each plan elevation, splash-block/lateral connection detail.
- **1.05 Quality Assurance:** Installer with a truck-mounted seamless machine and 3 years' experience.
- **1.06 Delivery, Storage, and Handling:** Roll-form on site; protect finished coil from scratching.
- **1.07 Warranty:** Coil finish 20-year; Contractor's 2-year workmanship (no leaks at outlets/miters).

Part 2 — Products

- **2.01 Manufacturers:** Coil: Amerimax (Euramax); Berger; Spectra Metals (Gutter Supply); Englert; or approved equal.
- **2.02 Materials:** 6-in K-style seamless aluminum, 0.027 in (0.032 in at runs > 40 ft or where guards are added), baked-enamel finish; 3 × 4-in corrugated rectangular downspouts, 0.019 in; hidden hangers with screws at ≤ 24 in o.c. (16 in o.c. where the roof is 8:12 and the eave is long); aluminum miters and outlets riveted and sealed.
- **2.03 Performance Criteria:** Sized for the 8:12 roof plane areas using SMACNA Table 1-2 for the Atlanta 5-minute rainfall intensity (≈ 8 in/hr, 10-yr — VERIFY); slope 1/16 in per ft to outlets; at least two downspouts per gutter run > 35 ft; discharge to splash blocks onto graded swales sloping away, or to solid 4-in PVC laterals to the lot drainage system where the lot grading plan requires.
- **2.04 Finishes:** Gutters and downspouts match trim color (warm white) on all schemes unless the Architect elects roof-matching color at the fascia.
- **2.05 Accessories:** Downspout straps at ≤ 6 ft; splash blocks (precast concrete) or lateral adaptors; optional micro-mesh gutter guards (stainless mesh, aluminum frame) as an HOA maintenance upgrade; 07 92 00 sealant at miters.

Part 3 — Execution

- **3.01 Examination:** Verify fascia is straight and drip edge extends into the gutter; verify downspout locations do not discharge onto walks, the zero-step landing, or a neighbor's lot.
- **3.02 Preparation:** Snap the slope line; set outlet positions at low points; pre-punch outlets.
- **3.03 Installation:** Install per SMACNA with the front lip below the roof plane extension; fasten hangers into rafter tails/sub-fascia; seal all joints with gutter sealant; strap downspouts plumb, 1 in off the siding on standoffs, terminating with an elbow to a splash block or into the lateral with a debris cleanout.
- **3.04 Field Quality Control:** Hose test each run for ponding (< 1/8 in standing water) and leaks before final painting.
- **3.05 Cleaning and Protection:** Remove sealant smears and debris; the HOA maintenance plan shall include semi-annual cleaning.

> ***[REVIEW REQUIRED]*** *Sizing rainfall intensity and the discharge method (splash block vs. lateral) depend on the civil engineer's lot grading and stormwater plan.*

Section 07 92 00 — Joint Sealants

Part 1 — General

- **1.01 Section Includes:** Exterior sealants at siding-to-trim, trim-to-masonry, window and door perimeters, penetrations, and flashings; horizontal traffic sealant at slab isolation joints and the zero-step landing joint; interior sanitary sealants at tile, tubs, showers, countertops; acoustical/air-seal sealants (with 07 27 00); interior painter's caulk.
- **1.02 Related Sections:** 03 30 00; 04 20 00 / 04 40 00; 07 27 00; 07 46 46; 08 series; 09 30 13; 12 36 00; 22 40 00.
- **1.03 References:** ASTM C920 (elastomeric joint sealants); ASTM C834 (latex sealants); ASTM C1193 (guide for use of joint sealants); ASTM C1330 (backer rod); ASTM C1521 (field adhesion); ASTM C1311 (solvent-release sealants — not used); SWRI Sealants Validation Program; siding manufacturer's sealant requirements (paintable, ASTM C920 Class 25 minimum, permanently flexible).
- **1.04 Submittals:** Product data and color charts; joint schedule (location, sealant type, backer, primer); adhesion test log.
- **1.05 Quality Assurance:** Installer with 3 years' experience; field adhesion tests (ASTM C1521 hand-pull) at the mock-up house for each exterior substrate pairing.
- **1.06 Delivery, Storage, and Handling:** Store 40–80 degrees F; use before the expiration date.
- **1.07 Warranty:** Manufacturer's 20-year (silicone) / 5-year (polyurethane) material warranty; Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** Sika (Sikaflex); Tremco (Dymonic 100, Spectrem); Dow (DOWSIL 790/795/786); Pecora (890NST, 898NST sanitary); GE Silicones; or approved equal.
- **2.02 Materials:** Exterior siding/trim/masonry: ASTM C920 Type S, Grade NS, Class 25 (or 35), Use NT/M/A/O, paintable polyurethane or hybrid (e.g., Dymonic 100, Sikaflex-15LM) — or, where a non-paintable joint is acceptable, neutral-cure silicone Class 50 in matching color; horizontal slab and landing joints: ASTM C920 Type S, Grade P, Class 25, Use T self-leveling polyurethane; sanitary: ASTM C920 mildew-resistant 100 percent silicone, Use NT; interior trim: ASTM C834 siliconized acrylic latex; acoustical/air-seal: non-hardening synthetic rubber; backer rod ASTM C1330 Type C (closed-cell or bicellular) sized 25 percent larger than the joint.
- **2.03 Performance Criteria:** Joint width 1/4 in minimum at siding-to-trim where sealed (1/8 in field gap per 07 46 46 is sealant-filled without backer); width-to-depth 2:1, depth 1/4–1/2 in; ± 25 percent movement (Class 25) at siding/trim, ± 50 percent at masonry-to-siding transitions; no three-sided adhesion (bond breaker at shallow joints).
- **2.04 Finishes:** Exterior colors matched to adjacent siding/trim/masonry per scheme (custom color where standard colors do not match); sanitary sealant white or clear; interior trim caulk painted.
- **2.05 Accessories:** Primers per the manufacturer's substrate chart (fiber cement cut edges, masonry, PVC); masking tape; bond-breaker tape.

Part 3 — Execution

- **3.01 Examination:** Verify joints are dry, clean, and within width tolerance; verify backer rod is required (joints > 3/8 in).
- **3.02 Preparation:** Clean per ASTM C1193; prime where required; install backer rod or bond-breaker tape; mask adjacent finishes.
- **3.03 Installation:** Tool concave with a wet tool within the working time; slab joints: fill to 1/8 in below the surface with self-leveling sealant over backer; sanitary joints at all changes of plane in tile (09 30 13) and at fixture-to-counter joints; do not seal weeps, flashed butt joints, or the bottom edge of lap siding.
- **3.04 Field Quality Control:** ASTM C1521 hand-pull at 5 locations on the mock-up house per substrate; re-test any failed pairing after primer change.
- **3.05 Cleaning and Protection:** Remove excess immediately with the manufacturer's solvent; protect from traffic until cured (24–72 hours).

> ***[REVIEW REQUIRED]*** *Sealant types and colors are generic pending the Architect's joint schedule; confirm siding-manufacturer sealant requirements for the factory-finish warranty.*

Division 08 — Openings

Section 08 14 00 — Wood Doors (with 08 16 13 Fiberglass Entry Doors)

Part 1 — General

- **1.01 Section Includes:** Fiberglass (basis of design) or wood exterior entry doors, 3'-0" × 6'-8" × 1-3/4 in, at front, rear/patio, and garage-to-house locations, with zero-threshold sill systems; interior molded or flush wood doors 3'-0" × 6'-8" (32-in minimum clear opening) with lever hardware; door frames, weatherstripping, and hardware.
- **1.02 Related Sections:** 03 30 00 (landing elevation); 06 10 00 (rough openings, garage separation); 07 27 00 (sill pans, flashing); 07 92 00; 09 91 13 / 09 91 23 (finishes); 26 50 00 (porch light at entry).
- **1.03 References:** 2024 IRC (GA 2026) R311.2 (egress door: 32-in clear, 78-in height), R311.3.1 (landing at the required egress door **not more than 1-1/2 in below the top of the threshold; 7-3/4 in permitted where the door does not swing over the landing** — the design uses ≤ 1/2 in), R302.5.1 (garage-to-house door: solid wood 1-3/8 in, solid/honeycomb steel, or 20-minute rated; **self-closing and self-latching**), R308.4 (safety glazing in doors and sidelites), R609 (exterior doors: testing/labeling); ICC A117.1-2017 §404.2.3 (32-in clear), §404.2.4 (thresholds ≤ 1/2 in, beveled 1:2), §404.2.6 (lever hardware 34–48 in AFF), §404.2.8 (≤ 5 lbf interior opening force); AAMA/WDMA/CSA 101/I.S.2/A440-22 (NAFS) for exterior doors; WDMA I.S. 1A (interior architectural wood flush doors) / I.S. 6A (molded); ANSI/BHMA A156.2 (locks), A156.1 (hinges), A156.4 (closers); NFRC 100/200; ENERGY STAR Version 7.0 (doors).
- **1.04 Submittals:** Door schedule; product data with NFRC labels and DP rating; threshold/sill pan detail at the zero-step landing; hardware schedule with finish; interior door style sample.
- **1.05 Quality Assurance:** Installer trained by the door manufacturer; the first entry door installed is the mock-up (water test the sill).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store doors upright, covered, in a conditioned space; fiberglass entry doors limited lifetime (glass 20-year, finish 10-year); interior doors 5-year; hardware 10-year mechanical / 5-year finish; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Entry: Therma-Tru (Fiber-Classic / Smooth-Star), Masonite (Belleville / Barrington), ProVia (Signet), JELD-WEN (Smooth-Pro), Pella; interior: Masonite, JELD-WEN, Steves & Sons; sills: Endura Products, Pemko; hardware: Schlage, Kwikset (Halo/SmartKey), Emtek; or approved equal.
- **2.02 Materials:** Entry doors: fiberglass skins (smooth for schemes 1 and 2; fir-grain for scheme 3 stain-grade look), polyurethane foam core, composite (rot-proof) bottom rail and jambs, factory-hung in a composite/primed frame; glazing: tempered, low-E insulated, decorative or clear per elevations with sidelite where shown; **threshold: low-profile "ADA" sill, ≤ 1/2 in total height with 1:2 bevel, with a multi-point or adjustable sill cap, integral sill pan/end dams, and a high-performance door bottom (double bulb)** — the covered 6-ft porch and sloped landing are the primary water defense; rear door: same, or 6'-0" sliding/hinged patio door with a low-profile track (≤ 1/2 in) per plan; garage-to-house: 20-minute fire-rated fiberglass or 1-3/8-in solid-core wood, with self-closing hinges and a self-latching lockset (R302.5.1), weatherstripped; interior: 1-3/8-in molded hollow-core (solid-core at bedrooms and baths for acoustics) 2-panel shaker, primed; pocket doors (with lever-style edge pulls) where plans show, no barn doors on accessible routes.
- **2.03 Performance Criteria:** Exterior doors NAFS Performance Grade 30 minimum (DP 30, VERIFY against the adopted wind speed); NFRC U ≤ 0.20 (opaque) / ≤ 0.25 (glazed) meeting ENERGY STAR v7.0 door criteria (VERIFY current values); air leakage ≤ 0.5 cfm/ft² for swinging doors (IECC R402.4.3); clear opening 32 in minimum at every door; lever hardware; 5-lb maximum interior opening force; no thresholds at interior doors (09 65 00 continuous flooring).
- **2.04 Finishes:** Entry door exterior: scheme 1 deep charcoal (illustrative #3B3F44), scheme 2 near-black (#2B2B2B) or natural stain, scheme 3 deep green (#2F4F3E) — as selected; interior face and all interior doors semi-gloss white (09 91 23); hardware satin nickel or matte black per scheme.
- **2.05 Accessories:** Ball-bearing hinges 4 × 4 in (3 per door; 4 at 8-ft doors); privacy sets at baths/bedrooms, passage elsewhere; door stops; peephole/door viewer at 43 and 60 in (two heights) or a video doorbell; smart deadbolt (keypad) option; weatherstripping (Q-lon) and corner seals; sill pans (07 27 00).

Part 3 — Execution

- **3.01 Examination:** Verify rough opening size and square; **verify the landing is within 1/2 in of the interior floor and slopes away 2 percent** before setting the sill.
- **3.02 Preparation:** Install sill pan with end dams and back dam sealed to the subfloor/slab; shim points at hinges and strike.
- **3.03 Installation:** Set entry doors per ASTM E2112 and the manufacturer: bed the sill in two beads of sealant over the pan, plumb and square with 1/8-in even reveal, foam (low-expansion) the perimeter, flash head and jambs to the WRB; adjust sill cap for a compressed sweep seal; interior doors plumb with 1/8-in reveals, hardware at 36 in centerline (levers 34–48 in AFF), strikes adjusted for self-latching at the garage door.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Spray test the entry sill at the mock-up house (AAMA 502 field test optional); verify each garage door self-closes and latches from a 6-in open position; protect doors with cardboard until final

paint; remove labels except NFRC/fire labels.

[REVIEW REQUIRED] *Life-safety adjacent (egress, garage separation, safety glazing) and the zero-threshold sill (well inside the R311.3.1 limits but demanding a drawn water-management detail) — the Architect shall detail and the AHJ shall confirm.*

Section 08 36 13 — Sectional Doors (Carriage-Style Insulated Garage Door)

Part 1 — General

- **1.01 Section Includes:** One 16'-0" × 7'-0" insulated steel sectional overhead door per cottage, carriage-house ("long-panel" recessed) design with optional top-row windows and decorative hardware; tracks, torsion springs, weatherseals; belt-drive opener with photo-eyes and battery backup.
- **1.02 Related Sections:** 03 30 00 (garage apron, zero-step slab); 06 10 00 (header, jamb backing); 07 46 46 (trim); 07 62 00 (head flashing); 26 00 00 (opener circuit, not in this outline).
- **1.03 References:** 2024 IRC (GA 2026) R609 (garage doors tested per ASTM E330 or ANSI/DASMA 108 and labeled), R309 (garage floor slope to the door or to a drain); ANSI/DASMA 102 (sectional door specifications), ANSI/DASMA 108 (wind load), DASMA 116 (impact — not required); UL 325 (openers: entrapment protection, photo-eyes); ASTM A653 (galvanized steel); ASTM C1363 (thermal; DASMA TDS 163 R-value method).
- **1.04 Submittals:** Product data with wind-load label, R-value per DASMA TDS 163, window and hardware options, color chips; opener data with UL 325 listing.
- **1.05 Quality Assurance:** Installer IDEA-certified (Institute of Door Dealer Education and Accreditation) or manufacturer-authorized; spring counterbalance rated 25,000 cycles minimum (HOA durability).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver after siding and painting; store sections flat; door limited lifetime (steel skin) / 10-year (hardware, insulation delamination) / 3-year (paint finish); opener 10-year motor, 5-year parts (belt lifetime); Contractor 1-year (springs adjusted).

Part 2 — Products

- **2.01 Manufacturers:** Clopay (Coachman); Amarr (Classica); Overhead Door (Courtyard / Carriage House 5900 series); Wayne Dalton (9700 series); C.H.I. Overhead Doors (5983); openers: LiftMaster (8550W/87504), Genie, Chamberlain; or approved equal.
- **2.02 Materials:** Three-layer door: 24–27-ga hot-dip galvanized steel skins (embossed carriage design, ~2-in polyurethane foam core), 1-3/8 to 2-in section thickness, steel end stiles; **R-value 12 minimum (R-16+ preferred) per DASMA TDS 163**; 2-in galvanized track, 14-ga brackets, nylon rollers with sealed ball bearings (quiet operation for 55+ homes), torsion springs, bottom aluminum retainer with U-shaped vinyl seal, vinyl jamb weatherseal; optional insulated (double-glazed) top-row windows with SDL grilles matching the house windows; decorative black handles/hinges per scheme.
- **2.03 Performance Criteria:** Wind-load rated per ANSI/DASMA 108 for the site's design pressure (115 mph Exposure B, 1-story — VERIFY; typically ± 20–25 psf) and labeled; air infiltration seal on all four sides; opener 3/4-hp DC belt drive with battery backup, photo-eyes at 6 in, auto-reverse (UL 325), soft start/stop, Wi-Fi/MyQ-type app control, wall console with lock and light, two remotes, exterior keypad at 48 in AFF; door opening force (manual) ≤ 25 lb with springs balanced.
- **2.04 Finishes:** Factory baked-on polyester (or PVDF where offered) in scheme 1 warm white, scheme 2 white or sage (VERIFY availability; otherwise paint per 09 91 13 with the manufacturer's approved system), scheme 3 warm white or almond — as selected.
- **2.05 Accessories:** Perimeter PVC stop molding with seal; exterior trim (07 46 46); safety pull-cord, emergency release; optional "Torquemaster"-type enclosed springs; low-headroom kit only where trusses require.

Part 3 — Execution

- **3.01 Examination:** Verify opening 16'-0" × 7'-0" ± 1/4 in, jambs plumb, headroom ≥ 12 in (or per opener), backroom ≥ 8 ft, and a level slab at the door line consistent with the zero-step apron (garage slab may slope to the door per R309.1 — VERIFY the apron detail).
- **3.02 Preparation:** Install 2×6 jamb and header backing (06 10 00); install center bearing plate backing.
- **3.03 Installation:** Per DASMA and the manufacturer's instructions; sections level, tracks plumb, springs wound and balanced so the door stays in place at any height; opener rail level, force settings and travel limits set, photo-eyes aligned, wall console mounted per the manufacturer's UL 325 instructions (typically ≥ 5 ft AFF, out of children's reach) with the keypad, remotes, and app control providing seated-reach operation (A117.1 §308 — VERIFY the AHJ's position on a lower console); weatherseal compressed 1/2 in.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Verify the UL 325 reversal with a 2×4 test and the photo-eye test; verify the wind-load label matches the submittal; wipe panels; leave the operating manual and spring-cycle rating with the HOA closing package; no painting of weatherseals or rollers.

> ***[REVIEW REQUIRED]*** *Wind-load design pressure and label class must be confirmed against the adopted wind speed; door color availability by scheme is unverified.*

Section 08 53 13 — Vinyl Windows (Fiberglass Alternate — 08 54 13)

Part 1 — General

- **1.01 Section Includes:** Single-hung or double-hung vinyl (uPVC) windows (fiberglass or composite as an approved alternate), 1-over-1 (schemes 2 and 3) and 2-over-2 simulated-divided-lite (scheme 1), with insulating low-E glass, integral nailing flanges, screens, and hardware; fixed and awning units where shown; mulled assemblies; egress-sized bedroom units.
- **1.02 Related Sections:** 06 10 00 (rough openings, headers); 07 27 00 (sill pans, flashing, WRB laps); 07 46 46 (trim); 07 92 00; 09 91 13 (exterior trim paint, not the window); 12 24 00 window shades (not in this outline).
- **1.03 References:** 2024 IRC (GA 2026) R310 (emergency escape and rescue openings: 5.7 SF net clear, 24-in height, 20-in width, sill ≤ 44 in AFF, at every sleeping room), R308.4 (safety glazing near doors, tubs/showers, and large low panes), R312.2 (window fall protection — not triggered at one story), R609 (exterior windows: NAFS performance and labeling, installation per the manufacturer and flashing per R703.4); 2015 IECC (GA) Table R402.1.2 (U-0.35 / SHGC-0.27) and §R402.4.3 (air leakage ≤ 0.3 cfm/ft²); AAMA/WDMA/CSA 101/I.S.2/A440-22 (NAFS-22); AAMA 303 (vinyl profiles); NFRC 100/200/400; ENERGY STAR Version 7.0 (Windows, Doors, and Skylights); ASTM E2112 (installation); ICC A117.1-2017 §309 (operable parts ≤ 5 lbf, 15–48 in reach).
- **1.04 Submittals:** Window schedule with NFRC U/SHGC/VT, NAFS PG rating, egress net-clear-opening calculation for each bedroom unit, SDL bar profile, exterior/interior color samples, screen and hardware samples; installation detail set (sill pan, jamb, head) coordinated with 07 27 00.
- **1.05 Quality Assurance:** Windows from a single manufacturer; installer trained (AAMA InstallationMasters or manufacturer program); mock-up: one bedroom window in the envelope mock-up house, water-tested (AAMA 502, optional).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store upright, off the ground, covered, out of direct sun; vinyl frame limited lifetime, insulated glass 20-year (seal failure), hardware 10-year, exterior color (dark laminates/paint) 10-year, transferable to subsequent owners; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Vinyl: Pella (250 Series), Andersen (100 Series Fibrex composite), MI Windows and Doors (V3000 / EnergyCore), Cornerstone/Ply Gem (1500 Series), JELD-WEN (V-4500), Milgard (Trinsic); fiberglass alternate: Marvin (Elevate/Essential), Pella (Impervia), Milgard (Ultra); or approved equal.
- **2.02 Materials:** Multi-chamber extruded uPVC (AAMA 303) with fusion-welded corners, or fiberglass/composite alternate; single-hung (double-hung at the Owner's option) with tilt-in sash, block-and-tackle balances, cam lock(s) at ≤ 5 lbf; glazing: double-pane, 3/4-in IGU, low-E (spectrally selective, e.g., Cardinal LoE-366 or equal), argon fill, warm-edge spacer, tempered where R308.4 applies; **SDL grilles: 7/8-in exterior/interior applied bars with a spacer bar in the airspace (2-over-2 on scheme 1; 1-over-1 no grilles on schemes 2 and 3)**; full or half fiberglass-mesh screens (aluminum frames); integral nailing fin, J-channel not used (trimmed per 07 46 46).
- **2.03 Performance Criteria:** NAFS Performance Class R or LC, PG 30 minimum (DP 30 — VERIFY with the adopted wind speed and the sizes on the elevations); **NFRC U-factor** ≤ **0.28 and SHGC** ≤ **0.23** as the design basis — this meets ENERGY STAR v7.0 for both the South-Central and Southern zones (Gwinnett County's ENERGY STAR zone to be confirmed on the ENERGY STAR climate-zone finder) and exceeds the Georgia prescriptive U-0.35 / SHGC-0.27; VT ≥ 0.45; air leakage ≤ 0.3 cfm/ft² (IECC R402.4.3); egress at bedrooms per R310 (a 3'-0" × 5'-0" single-hung typically provides ~5.0–5.7 SF; confirm each unit — casement or a larger unit may be needed); sill heights 30–36 in AFF for seated views, with locks within a 15–48 in reach.
- **2.04 Finishes:** Exterior: white (schemes 1 and 2) or bronze/black laminate (scheme 3), as selected; interior white; hardware white or matching; screen frames match exterior.
- **2.05 Accessories:** Mull kits with structural reinforcement for paired units; factory-applied fin sealant; snap-in grilles are not acceptable (SDL specified); window-opening control devices not required (one story).

Part 3 — Execution

- **3.01 Examination:** Verify rough opening +1/2 in width / +1/4 in height (or per manufacturer), square within 1/8 in, and sill pan/WRB prepared per 07 27 00.
- **3.02 Preparation:** Apply sealant at the fin (jambs and head, not the sill unless the manufacturer requires a weep gap); position shims at 1/4 points.
- **3.03 Installation:** Per ASTM E2112 Method A1 (fin over WRB with self-adhered jamb and head flashing, WRB lapped over the head flashing) and R609; fasten fins at 8–12 in o.c.; foam (low-expansion) the perimeter; install exterior trim with a 1/8-in sealed gap to the frame; verify sash operates and locks with ≤ 5 lbf.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Egress net-clear-opening verified on one installed unit per type; NFRC labels left on until inspection, then removed; protect from mortar and paint; clean glass with non-abrasive cleaner; replace scratched IGUs.

  [REVIEW REQUIRED] *Life-safety adjacent (egress and safety glazing); PG rating, ENERGY STAR zone, and the egress sizing of every bedroom window require the Architect's schedule.*

Division 09 — Finishes

Section 09 29 00 — Gypsum Board

Part 1 — General

- **1.01 Section Includes:** Interior gypsum board on walls and ceilings; moisture- and mold-resistant board at baths, laundry, and kitchen wet walls; garage/house separation board (fire-taped); cement board or glass-mat backer at tile; Level 4 finish (Level 5 at the great room ceiling where lit by clerestory/wash light, optional); corner beads and trims; the ceiling-plane air barrier.
- **1.02 Related Sections:** 06 10 00 (framing, blocking, fireblocking); 07 21 00; 07 27 00 (airtight drywall); 09 30 13 (tile backer); 09 91 23 (paint); 10 28 13 (grab bars fasten through the board into blocking).
- **1.03 References:** 2024 IRC (GA 2026) R702.3 (gypsum board), Table R702.3.5 (application — 1/2-in board on 24-in o.c. ceilings only with restrictions; see 2.02), R702.4 (tile backers at tub/shower walls: cement, fiber-cement, or glass-mat gypsum), R302.6 (garage separation: 1/2-in gypsum on the garage side of walls and ceilings; 5/8-in Type X where habitable space above — not applicable), R302.11 (fireblocking); ASTM C1396 (gypsum board), C1178 (glass-mat gypsum), C1325 (cementitious backer), C840 (application and finishing), C475 (joint materials), C1002 (screws), C1047 (trims); GA-214 (Levels of Gypsum Board Finish); ASTM D3273 (mold resistance, score 10).
- **1.04 Submittals:** Product data; finish-level plan; garage separation detail; texture sample (if any).
- **1.05 Quality Assurance:** Installer with 5 years' experience; sample room (the first cottage's primary bath and great room) approved for finish level before proceeding.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver after dry-in; store flat in conditioned space, off the floor; do not install above 90 percent RH; Contractor 2-year (no nail pops or cracked joints, one repair visit at 11 months).

Part 2 — Products

- **2.01 Manufacturers:** Georgia-Pacific Gypsum (ToughRock, DensArmor Plus, DensShield — Atlanta, GA); USG (Sheetrock, Mold Tough, Durock); National Gypsum (Gold Bond, XP, PermaBase); CertainTeed; or approved equal.
- **2.02 Materials:** Walls: 1/2-in ASTM C1396 regular board; ceilings on 24-in o.c. trusses: **5/8-in regular or 1/2-in sag-resistant "ceiling board"** (Table R702.3.5 — 1/2-in regular board is not permitted at 24 in o.c. under water-based texture); baths, laundry, kitchen sink wall: 1/2-in moisture/mold-resistant (ASTM C1396 + D3273 score 10; paperless glass-mat preferred in showers' dry zones); tile backer at showers and tub walls: 1/2-in cement board (C1325) or glass-mat water-resistant gypsum backer (C1178, with bonded waterproofing per 09 30 13); garage side of separation walls/ceiling: 1/2-in regular (5/8-in Type X optional for the HOA's insurance benefit), joints taped (fire-taped); joint compound setting-type at first coat, ready-mix finish; paper-faced metal or vinyl corner bead; bullnose at the Owner's option (not at tile or door casings).
- **2.03 Performance Criteria:** Level 4 finish per GA-214 at all painted surfaces (Level 5 optional where noted); ceiling board fastened at 12 in o.c. field, 7 in perimeter (screws 1-1/4 in); walls 16 in o.c.; no board within 1/2 in of the slab (sealed gap, 07 27 00); air barrier: gypsum ceiling sealed to top plates and around all penetrations (IECC Table R402.4.1.1).
- **2.04 Finishes:** Smooth (no texture) walls and ceilings — a light knockdown texture on ceilings only at the Owner's option; no popcorn; painted per 09 91 23.
- **2.05 Accessories:** Control joints (C1047) at ≤ 30 ft in long ceilings; resilient channel not required; acoustical sealant at bedroom/bath partitions (optional); access panels at sprinkler valves/plumbing (11-in × 14-in, paintable).

Part 3 — Execution

- **3.01 Examination:** Verify framing is dry (≤ 19 percent MC), blocking is complete (photo log per 06 10 00), the pre-drywall energy/insulation inspection has passed, and rough-in inspections are signed.
- **3.02 Preparation:** Install backer at tile areas first; install ceiling-plane air-sealing gaskets at top plates; mark grab-bar blocking lines on the framing edge for the tile setter and accessory installer.
- **3.03 Installation:** Per ASTM C840 and GA-216: ceilings first, perpendicular to trusses with floating interior angles; walls horizontal, joints staggered; screw-attach (no nails) in ceilings; garage separation continuous with taped joints and fire-rated sealant at penetrations; finish Level 4 (three coats over tape, sanded); cement board joints with alkali-resistant mesh and

thin-set.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect the finish under a raking light at 45 degrees (side-lit) before priming; verify garage separation continuity above the ceiling line; vacuum sanding dust; protect tile backer from water until the membrane is applied.

[REVIEW REQUIRED] *Life-safety adjacent (garage separation, fireblocking); ceiling board type on 24-in o.c. trusses and the Level 5 extent to be confirmed by the Architect.*

Section 09 30 13 — Ceramic Tiling

Part 1 — General

- **1.01 Section Includes:** Porcelain floor tile at baths and laundry; ceramic/porcelain wall tile at curbless (roll-in) showers to the ceiling and at tub surrounds where a tub option is elected; kitchen backsplash tile; sloped mortar beds, bonded waterproofing, linear drains (drain body by 22 40 00), grout, and transitions flush with LVP (no thresholds).
- **1.02 Related Sections:** 03 30 00 (1-1/2-in slab recess at showers); 07 92 00 (sealant at changes of plane); 09 29 00 (backer); 09 65 00 (flush transition); 10 28 13 (grab bars through tile into blocking); 22 40 00 (linear drain, valve, hand-held shower).
- **1.03 References:** ANSI A108 / A118 / A136.1 (installation and materials), ANSI A137.1 (tile), ANSI A326.3 (DCOF); TCNA Handbook 2025 — methods **B422C (curbless shower, bonded waterproofing over slab recess)**, F125A (crack isolation), W244 (backer-board walls), EJ171 (movement joints); ASTM C373, C1028 (superseded — use A326.3), C1027 (abrasion); ICC A117.1-2017 §608.2.2 (roll-in shower 30 × 60 in minimum; no curb; ≤ 1/2-in beveled entry) and §608.6 (hand-held shower), §303 (changes in level); 2024 IPC (GA 2026) §417.5.2 (shower floor slope 1/4 in per ft to the drain — **VERIFY the section number in the 2024 edition**) and §417.4 (shower compartment 900 in² minimum).
- **1.04 Submittals:** Tile samples (two per type/color), grout and sealant color charts, membrane and mortar data, shower layout with drain position and slope, DCOF report for floor tile.
- **1.05 Quality Assurance:** Tile setter CTEF Certified Tile Installer (or 5 years' documented experience); mock-up: the first cottage's primary shower flood-tested for 24 hours before tile.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver in original cartons with dye lot; store dry above 50 degrees F; tile manufacturer 10-year; membrane manufacturer system warranty 10-year (where the setting-system manufacturer supplies all components); Contractor 2-year (no cracking, debonding, grout failure, or leaks).

Part 2 — Products

- **2.01 Manufacturers:** Tile: Daltile (Mohawk), Florida Tile, Crossville, MSI, Emser; setting systems: Schluter Systems (Kerdi/Kerdi-Line), Laticrete (Hydro Ban), Custom Building Products (RedGard, Prolite), Mapei; or approved equal.
- **2.02 Materials:** Bath floors: porcelain, ANSI A137.1 P1 (impervious, ≤ 0.5 percent absorption), 12 × 24 in matte or 2 × 2 in mosaic; shower floors: 2 × 2 in porcelain mosaic (or 1 × 1 in) for slope conformance; shower walls: 12 × 24 in porcelain or 3 × 6 / 4 × 12 in ceramic (A137.1 W or P) to the ceiling; backsplash: 3 × 6 in ceramic (or as selected); mortar: ANSI A118.15 (large format) / A118.4; grout: A118.7 high-performance cementitious, or A118.3 epoxy at shower floors; bonded waterproofing A118.10 (sheet or liquid, with pre-formed corners and drain integration); crack-isolation A118.12 over the slab at bath floors; sloped mortar bed A108.1A (or pre-sloped foam tray with linear-drain kit); linear drain: stainless, 32–48 in, wall-side or entry-side placement per plan, ASME A112.6.3 / A112.18.2 (VERIFY drain listing) — provided under 22 40 00.
- **2.03 Performance Criteria:** DCOF ≥ 0.42 (A326.3, interior wet) at all bath and shower floors; shower floor slope 1/4 in per ft (≥ 2 percent) to the drain within a 36 × 60 in minimum roll-in footprint; **shower entry: zero curb, ≤ 1/4 in vertical or ≤ 1/2 in beveled change in level (A117.1 §303)**; bath floor tile finished flush (± 1/8 in) with adjacent LVP by setting the tile assembly in the recessed slab; movement joints at the perimeter and at changes of plane (07 92 00) per EJ171; 95 percent mortar coverage in wet areas; lippage ≤ 1/32 in (rectified) per A108.02.
- **2.04 Finishes:** Colors as selected per scheme from the manufacturer's full range (warm gray/greige floors; white or soft-white walls; sanded grout color matched); matte finishes only on floors.
- **2.05 Accessories:** Schluter-type metal edge trim at outside corners and tile-to-drywall edges; recessed niche (pre-formed, waterproofed) and a 17–19-in-high built-in bench or fold-down seat (10 28 13) in every shower; grab-bar locations pre-drilled with diamond bits into blocking; pre-formed curbless shower tray systems permitted as an alternate to site-sloped beds.

Part 3 — Execution

- **3.01 Examination:** Verify the slab recess (1-1/2 in) and drain rough-in elevation allow the full slope to the drain with the tile finishing flush at the entry; verify backer installation, blocking, and the slab's moisture (ASTM F2170 RH or per the membrane manufacturer) and flatness (1/8 in in 10 ft for large format).
- **3.02 Preparation:** Form and cure the sloped bed (or set the pre-sloped tray); install the bonded waterproofing continuous from the drain flange up the walls to 72 in minimum (full height in the roll-in zone), with corners and seams per the manufacturer;

flood test 24 hours (ASTM D5957); install crack-isolation at bath floors.

- **3.03 Installation:** Per ANSI A108 and the TCNA method: back-butter large format; joint width 1/8 in (3/16 in at mosaics); align shower-wall courses to the floor grid; hold sealant (not grout) at all changes of plane and at the tile/LVP joint if not flush; grout after 24 hours; seal cementitious grout after cure.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Water test the finished shower (fill to the entry line, verify drainage with no ponding > 1/8 in); lift one tile per shower floor to verify coverage; clean haze within 24 hours; protect floors with breathable covering until Substantial Completion.

__[REVIEW REQUIRED]__ Curbless-shower assembly, slab recess depth, drain type, and the flush LVP transition are interdependent details the Architect must draw; confirm IPC 2024 section numbers.

Section 09 65 00 — Resilient Flooring (Luxury Vinyl Plank)

Part 1 — General

- **1.01 Section Includes:** Rigid-core (SPC or WPC) luxury vinyl plank throughout living areas, kitchen, bedrooms, hall, closets, and den — continuous with no interior thresholds or transition strips at doorways; flush transitions to tile (09 30 13) and to the garage/entry door thresholds; resilient/wood base by 06 20 00.
- **1.02 Related Sections:** 03 30 00 (slab flatness, moisture, curing compound compatibility); 06 20 00 (base and shoe); 09 30 13; 08 14 00 (door undercuts, thresholds).
- **1.03 References:** ASTM F3261 (rigid-core resilient plank — Standard Specification); ASTM F710 (preparing concrete floors); ASTM F2170 (in-situ RH) and F1869 (calcium chloride); ASTM E648 (critical radiant flux, Class I ≥ 0.45 W/cm²) and E662 (smoke ≤ 450); ASTM D2047 (static COF ≥ 0.5, dry); ANSI/RFCI; FloorScore / CDPH Standard Method v1.2 (VOC); ICC A117.1-2017 §302 (firm, stable, slip-resistant), §303 (no change in level > 1/4 in); 2024 IRC (GA 2026) R311.3 (doorway floor levels).
- **1.04 Submittals:** Product data with wear-layer thickness, ASTM F3261 classification, locking-system run limits, FloorScore certificate, 3 full-length plank samples per color, layout plan with maximum run lengths and expansion provisions.
- **1.05 Quality Assurance:** Installer certified by the manufacturer; mock-up: the first cottage's great room/kitchen run reviewed for the flush tile transition and end-joint stagger.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Acclimate in the conditioned house 48 hours at 65–85 degrees F; store flat; manufacturer's limited lifetime residential wear warranty (25-year minimum), waterproof warranty; Contractor 2-year (no gapping, peaking, or hollow spots).

Part 2 — Products

- **2.01 Manufacturers:** Shaw Floors (Floorté / COREtec — Dalton, GA); Mohawk (SolidTech — Calhoun, GA); Mannington (Adura Max/Rigid); AHF Products (Armstrong Flooring); or approved equal.
- **2.02 Materials:** Rigid-core plank, ASTM F3261, 5.5–8 mm total thickness with attached acoustic pad, **wear layer ≥ 20 mil**, ceramic-bead or urethane finish, 7 × 48 in (or 9 × 60 in) planks, beveled micro-edge, angle-tap or drop-lock joint; waterproof core; phthalate-free, FloorScore certified.
- **2.03 Performance Criteria:** Class I (E648); SCOF ≥ 0.5 (D2047); indentation ≤ 0.010 in (ASTM F1914, 250-lb static load — 55+ walkers/wheelchairs); maximum continuous run per the manufacturer (typically 50–60 ft) — both plans fit within a single expansion zone so no doorway T-moldings are required; perimeter expansion 3/8 in concealed under base/shoe; **transitions: flush (≤ 1/8 in) to tile by slab recess, or a 1/4-in-maximum beveled reducer where flush is not achieved; no transition strips at interior doorways**; slab moisture ≤ 85 percent RH (F2170) or per the manufacturer; pH ≤ 9; flatness 3/16 in in 10 ft (F710).
- **2.04 Finishes:** Light-to-medium warm wood tones (e.g., "natural oak", "greige oak") as selected per scheme; low-gloss; high LRV contrast with base and door trim (white) for low-vision wayfinding.
- **2.05 Accessories:** Manufacturer's reducer/end-cap only at the garage door threshold and patio door; stainless or aluminum flush "Schluter Reno"-type profile at tile where needed; underlayment not required (attached pad); repair kit left with the HOA.

Part 3 — Execution

- **3.01 Examination:** Verify slab moisture, pH, flatness, and that curing compounds are removed or compatible; verify tile areas are set to the flush elevation; verify doors are hung so undercut clearance is 1/2 in over the finished floor.
- **3.02 Preparation:** Grind high spots and fill lows with cementitious patch (F710); sweep and vacuum; snap layout lines parallel to the longest wall or the main sightline from the entry; plan end-joint stagger ≥ 8 in.
- **3.03 Installation:** Floating per the manufacturer: 3/8-in perimeter gap at all walls, cabinets, and fixed objects (cabinets set on top of the floor is NOT permitted — cut around islands); run planks continuously through doorways; tap joints fully; cut neatly around the tile edge to a straight sealed joint; install base and shoe over the gap with no fasteners into the planks.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Walk test for hollow spots and peaking; check transitions with a straightedge (≤ 1/8 in); clean with the manufacturer's neutral cleaner; protect with breathable paper (no plastic) and hardboard at traffic paths until Substantial Completion.

[REVIEW REQUIRED] Product family (SPC vs. WPC), wear layer, and the flush-transition detail are Architect selections; confirm run limits against each plan's longest diagonal.

Section 09 91 13 — Exterior Painting (Three Color Schemes)

Part 1 — General

- **1.01 Section Includes:** Field painting of fiber-cement siding and trim (where not factory-finished), cellular PVC trim (where painted), porch ceilings, wood doors, exposed flashings/gutters where color-matched, and cedar staining; touch-up of factory-finished siding; color schedule for the three schemes.
- **1.02 Related Sections:** 06 20 00 (cedar, trim); 07 46 46 (factory finish alternative); 07 92 00 (paintable sealants); 08 14 00 / 08 36 13 (factory finishes); 09 91 23.
- **1.03 References:** MPI Architectural Painting Specification Manual (current edition) and MPI Approved Products List: MPI #6 (exterior latex primer, wood), MPI #3 (alkali-resistant primer, cementitious), MPI #15 (exterior latex, low sheen, G3–4) for siding, MPI #11 (exterior latex semi-gloss, G5) for trim and doors, MPI #33 (exterior semi-transparent stain) for cedar; ASTM D3359 (adhesion), D4442 (moisture); SSPC/PDCA P1–P3; 40 CFR 59 Subpart D (VOC — Georgia has no stricter AIM rule; use ≤ 100 g/L where available).
- **1.04 Submittals:** Color schedule per scheme (field, trim, door, porch ceiling, gutters, garage door), 8 × 10-in drawdowns of each color, product data, stain sample on the specified cedar; 4 × 8-ft mock-up per scheme on the model cottages.
- **1.05 Quality Assurance:** Painter with 5 years' exterior residential experience; one manufacturer for all coatings; mock-ups become the standard for sheen and coverage.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store 50–90 degrees F; no application below 50 degrees F, above 90 degrees F, or when rain is forecast within 24 hours; manufacturer's limited lifetime (premium acrylic); Contractor 3-year (no peeling, blistering, or chalking beyond ASTM D4214 No. 8).

Part 2 — Products

- **2.01 Manufacturers:** Sherwin-Williams (Duration / Emerald Exterior, WoodScapes stain); Benjamin Moore (Aura Exterior, Arborcoat); PPG (Manor Hall / Timeless, ProLuxe stain); or approved equal.
- **2.02 Materials:** 100 percent acrylic exterior latex, self-priming over factory-primed fiber cement (spot-prime cut edges with alkali-resistant acrylic primer); acrylic semi-gloss for trim, doors, and PVC (light-reflectance value ≥ 55 on PVC to limit heat distortion, per the trim manufacturer); cedar: penetrating oil/alkyd or waterborne semi-transparent stain, "natural cedar" tone, with a UV-blocking clear topcoat optional; galvanized/aluminum: acrylic bonding primer + finish where painted.
- **2.03 Performance Criteria:** Dry film thickness 1.5 mils per finish coat minimum, two finish coats (one on factory-finished touch-up); moisture in fiber cement ≤ 15 percent and cedar ≤ 15 percent before coating (D4442); sheen: siding low sheen (G3), trim/doors semi-gloss (G5), porch ceiling low sheen; door color coats on both faces; HOA repaint cycle planned at 8–10 years (factory ColorPlus extends to 15).
- **2.04 Finishes — color schedule (hex values are illustrative placeholders; final colors by the Architect from the manufacturers' full ranges):**
- **Scheme 1 "Warm White":** field board-and-batten warm white (#F3EFE6); trim, corners, fascia, and gutters soft white (#FAF8F3); front door deep charcoal (#3B3F44); porch ceiling pale sky (#DCE5E8); cedar posts/brackets natural stain; stone piers/water table gray-tan ledge; roof charcoal.
- **Scheme 2 "Sage":** field lap sage (#8E9C86); trim warm white (#F6F3EC); front door near-black (#2B2B2B) or natural stain; porch ceiling warm white; cedar painted trim color or natural (Architect's option); brick water table red-brown blend with buff mortar; roof weathered wood.
- **Scheme 3 "Clay/Beige":** field lap clay-beige (#CDB89A); trim cream (#F3EEE2); front door deep green (#2F4F3E); porch ceiling cream; cedar natural stain; stone buff/tan ledge; roof driftwood/brown.
- **2.05 Accessories:** Primers per substrate; caulk per 07 92 00 (painted); masking; touch-up kits (one quart of each color per cottage delivered to the HOA).

Part 3 — Execution

- **3.01 Examination:** Verify sealants are cured, siding clearances are correct, cedar is dry, and factory-finish touch-ups are complete; verify no visible flashing gaps remain to be "painted over".
- **3.02 Preparation:** Wash dust; sand PVC glossy spots; spot-prime nail heads, cut edges, and any bare substrate; back-prime replaced trim; mask masonry, windows, and stone.

- **3.03 Installation:** Brush-and-roll or spray-and-back-roll two finish coats; stain cedar two coats wiped, end grain saturated; paint doors on all six sides; do not paint weatherseals, weeps, labels, vents, or rollers; keep a wet edge on lap siding for full courses.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** DFT spot checks (Tooke gauge) at 3 locations per elevation on the mock-up; cross-hatch adhesion (D3359) on PVC and fiber cement at the mock-up; remove masking within 24 hours; clean overspray from stone, glass, and roofing; leave touch-up kits and the color schedule with the HOA.

  [REVIEW REQUIRED] *Colors are placeholders; MPI numbers should be confirmed against the current MPI manual; factory-finish (ColorPlus) vs. field paint is an Owner cost/maintenance decision that changes this section's scope.*

Section 09 91 23 — Interior Painting

Part 1 — General

- **1.01 Section Includes:** Priming and painting of gypsum board walls and ceilings, interior wood/MDF trim and doors, garage interior (walls and ceiling), closets, and interior side of exterior doors; low-VOC finishes; high-contrast trim for low-vision wayfinding.
- **1.02 Related Sections:** 06 20 00 (trim); 08 14 00 (doors); 09 29 00 (Level 4 substrate); 07 92 00 (interior caulk).
- **1.03 References:** MPI Architectural Painting Specification Manual: MPI #50 (latex primer sealer for gypsum), MPI #39 (latex primer for wood), MPI #53 (interior latex flat, G1) ceilings, MPI #52 (interior latex, G3 eggshell) walls, MPI #54 (interior latex semi-gloss, G5) or MPI #147 (waterborne alkyd, G5) trim and doors; MPI X-Green / GS-11 / CDPH v1.2 (VOC and emissions); GA-214 (finish level); ASTM D4828 (scrub resistance); IES RP-28 (lighting for older adults — contrast guidance).
- **1.04 Submittals:** Color schedule and 8 × 10-in drawdowns; product data with VOC content; sheen samples on the 09 29 00 sample room.
- **1.05 Quality Assurance:** Painter with 3 years' experience; sample room approval before proceeding; single manufacturer.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store 50–90 degrees F; apply at 60–85 degrees F with ventilation; manufacturer's limited lifetime; Contractor 1-year with a touch-up visit at 11 months.

Part 2 — Products

- **2.01 Manufacturers:** Sherwin-Williams (ProMar 200 Zero VOC / Duration Home / Emerald Urethane Trim); Benjamin Moore (Ultra Spec 500 / Regal Select / Advance); PPG (Speedhide Zero / Breakthrough); or approved equal.
- **2.02 Materials:** Gypsum primer (MPI #50, high-build where Level 4 flashing is a risk); walls: acrylic latex eggshell (G3), washable, ≥ 400 cycles scrub (D4828) — baths, laundry, and kitchen: mildew-resistant "kitchen and bath" satin; ceilings: flat (G1) ceiling paint; trim and doors: waterborne alkyd/urethane-modified semi-gloss (G5); garage: primer + one coat flat/eggshell (walls) and flat (ceiling); VOC ≤ 50 g/L (flat/eggshell), ≤ 100 g/L (semi-gloss).
- **2.03 Performance Criteria:** One primer coat plus two finish coats on new gypsum; two finish coats on trim over primer; DFT per MPI; **visual contrast: door/trim color vs. wall color LRV difference** ≥ **30 points** (low-vision aid); no sprayed "back-rolled-once" single-coat walls.
- **2.04 Finishes:** Walls warm off-white (illustrative "Alabaster"-type, LRV ~80); ceilings white flat; trim and doors bright white semi-gloss; one accent wall per plan optional; garage walls light gray.
- **2.05 Accessories:** Painter's caulk (07 92 00); masking; touch-up kit (one quart per color) per cottage.

Part 3 — Execution

- **3.01 Examination:** Verify Level 4 finish, dust-free surfaces, gypsum moisture < 12 percent, and that HVAC is operating for cure.
- **3.02 Preparation:** Sand and fill trim; caulk trim-to-wall joints after primer; remove hardware and cover plates; mask floors (09 65 00) with breathable protection.
- **3.03 Installation:** Prime, inspect under raking light, touch up defects, then two finish coats (spray-and-back-roll walls, brush trim); cut clean lines at ceiling; paint doors flat-laid or hung with hardware removed; paint closet interiors and behind fixtures.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect at 45-degree raking light; correct holidays and flashing; reinstall hardware; remove masking; ventilate 48 hours before flooring protection is removed.

  [REVIEW REQUIRED] *Generic colors and MPI numbers pending the Architect's interior finish schedule.*

Division 10 — Specialties

Section 10 14 00 — Signage (Monument Entry Sign)

Part 1 — General

- **1.01 Section Includes:** One ground-mounted monument entry sign, ≤ 32 SF of sign face, at the Arcado Road entrance: masonry base and cap matching the homes, dimensional letters "Cottages at Arcado Springs", ground-mounted shielded LED illumination, and footing. (Assigned to Division 10 rather than 13 34 00 Fabricated Engineered Structures because it is a signage assembly; the sign is permitted separately per the Lilburn checklist §4.q.)
- **1.02 Related Sections:** 03 30 00 (footing); 04 20 00 / 04 40 00 (brick or stone cladding by scheme — stone is the default at the entry); 26 50 00 (sign lighting); 32 93 00 (planting bed); 33 00 00 utilities/sight distance (not in this outline).
- **1.03 References:** Lilburn Zoning Ordinance 2023-603 sign regulations (subdivision/community entrance sign: permitted area, height, setback, illumination — **VERIFY the article and the 32-SF/height limits at the pre-application conference**; sign permit and possible sign easement per the Site Development Plan Review Checklist §4.q); Gwinnett County DOT driveway permit sight-distance triangle (no obstruction > 2.5 ft in the triangle — VERIFY); ASCE 7-22 (wind on the freestanding wall/sign — VERIFY the adopted edition); ASTM C1364 (cast stone); ASTM B209 / B26 (aluminum plate / castings); UL 48 and UL 879 (illuminated signs, where applicable); IES/IDA Model Lighting Ordinance (shielding).
- **1.04 Submittals:** Shop drawings with dimensions, area calculation, letter style/size, materials, footing, and illumination; masonry and cap samples; sign permit application drawings.
- **1.05 Quality Assurance:** Sign fabricator licensed in Georgia with 5 years' experience; masonry by the 04 series mason; footing by a Georgia PE if the sign exceeds 6 ft or 32 SF.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Protect letters and cap in crates; masonry per Division 04; letters 10-year (finish), LED fixtures 5-year, masonry 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** Letters: Gemini Incorporated (cast aluminum or flat-cut), Matthews Architectural Products, ASI Signage Innovations; HDU letters/panels: Coastal Enterprises (Precision Board), Sign Foam (SignArt); fabrication by a local sign contractor; or approved equal.
- **2.02 Materials:** Reinforced CMU core on a 3,000-psi footing 12 in below grade (frost) sized for wind; ledge/stacked stone (04 40 00) or brick (04 20 00) cladding; cast-stone cap (C1364) sloped with drip; sign face: 1/2-in cast-aluminum or 1/4-in flat-cut aluminum letters, 8–10-in cap height, pin-mounted with 1/2-in standoffs on a cast-stone or HDU panel (15-lb HDU, primed and painted); sign area ≤ 32 SF measured per the ordinance's method (VERIFY: face vs. panel); overall height ≤ 6 ft above grade (VERIFY).
- **2.03 Performance Criteria:** Located outside the Gwinnett DOT sight triangle and outside the public right-of-way, within (or adjacent to) the 10-ft landscape strip along Arcado Rd per the site plan (VERIFY setback); wind-resistant per ASCE 7 for the wall and letters; letter contrast ≥ 70 percent with the background; illumination by two ground-mounted, fully shielded LED floods ≤ 3000K (2700K preferred), aimed at the face only, with ≤ 0.1 fc spill at the property line and photocell/timer control (dark-sky); no internal illumination, no exposed bulbs, no changeable copy.
- **2.04 Finishes:** Letters: satin black or dark bronze (schemes 1–3 neutral); cap buff or gray per the default stone scheme; panel color matching the trim warm white.
- **2.05 Accessories:** Stainless pins and epoxy anchors; conduit sleeve through the footing; address numerals (4535–4541 range) on a separate address sign per the Gwinnett Fire Marshal (4-in minimum, contrasting); landscape bed with irrigation stub.

Part 3 — Execution

- **3.01 Examination:** Verify the surveyed location against the R/W, sight triangle, landscape strip, utilities (call 811), and the sign permit.
- **3.02 Preparation:** Excavate and pour the footing with anchor dowels and conduit; cure 7 days.
- **3.03 Installation:** Lay CMU core and cladding per Division 04; set the cap in full mortar with sealant joints; mount letters with a template, level, on standoffs; install and aim lights at night with the Architect present; plant the bed per 32 93 00.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Verify the sign area and height against the permit; night aiming check for glare from Arcado Rd; clean masonry (no acid) and letters; protect until planting is complete.

__[REVIEW REQUIRED]__ Sign area, height, setback, and illumination limits are unverified against the Lilburn sign article; a separate sign permit (and possibly a sign easement) is required.

Section 10 28 13 — Toilet Accessories (Bath Accessories and Grab Bars)

Part 1 — General

- **1.01 Section Includes:** Grab bars (installed, or "blocked-for" with locations recorded) at every toilet, shower, and tub; fold-down shower seats; shower slide bars (by 22 40 00); mirrors; robe hooks; towel bars/rings; toilet-paper holders; medicine cabinets; shower niches (09 30 13); the grab-bar blocking record for each cottage.
- **1.02 Related Sections:** 06 10 00 (blocking); 09 29 00; 09 30 13 (tile penetrations); 22 40 00 (fixtures, hand-held showers); 12 36 00 (vanity tops).
- **1.03 References:** ICC A117.1-2017 §604.5 (grab bars at water closets: side wall 42 in long, 12 in max from the rear wall; rear wall 36 in), §608.3 (roll-in shower: 24 in max from the entry along the back wall, full length of side walls per the variant), §608.4 (fold-down seat), §609 (grab bars: 1-1/4 to 2-in diameter, 1-1/2-in clearance, 33–36 in AFF, 250-lb load in any direction), §603.3 (mirrors: bottom edge ≤ 40 in AFF), §308 (reach ranges 15–48 in); ASTM A666 (stainless Type 304); ASTM F446 (grab bars) is withdrawn — structural loading is cited from A117.1 §609.8; the IRC has no grab-bar requirement (voluntary universal-design program).
- **1.04 Submittals:** Accessory schedule with mounting heights; product data; grab-bar layout per bath (one drawing per plan) showing the blocking zone; blocking photo log (06 10 00) delivered to the HOA.
- **1.05 Quality Assurance:** Installed by the finish carpenter; pull-test one installed grab bar per cottage to 250 lb.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver boxed after painting; grab bars 10-year (finish and structural), accessories 5-year, mirrors 5-year (silvering); Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Grab bars and seats: Bobrick, Bradley, ASI American Specialties (commercial-grade), Moen Home Care and Kohler (residential designer grade in matching finishes); mirrors/medicine cabinets: Kohler, Robern, Jensen; or approved equal.
- **2.02 Materials:** Grab bars 1-1/4- or 1-1/2-in diameter, 18-ga Type 304 stainless (satin or peened) or designer finishes matching the plumbing trim (matte black, satin nickel), concealed-flange mounting with #14 stainless screws into 06 10 00 blocking (no hollow-wall anchors); fold-down shower seat, phenolic or teak slats on a stainless frame, 250-lb rated, 17–19 in AFF; mirrors 1/4-in float glass (ASTM C1503) with polished edges or framed, bottom edge ≤ 40 in AFF; robe hooks ≤ 48 in AFF; toilet-paper holder 7–9 in in front of the toilet, 15–48 in AFF; towel bars at 48 in AFF.
- **2.03 Performance Criteria:** Every toilet: blocking for a 42-in side bar and a 36-in rear bar (install at closing at the Owner's election; at minimum install the side bar); every shower: blocking for the full back-wall and side-wall bars per §608.3, an 18-in vertical bar at the entry, and the fold-down seat; every tub (where the tub option is chosen): §607.4 layout; all bars 250-lb rated; contrast with wall tile where the buyer elects color bars.
- **2.04 Finishes:** Match the 22 40 00 trim finish per bath; satin stainless where designer finishes are unavailable.
- **2.05 Accessories:** Stainless mounting hardware; escutcheon sealant (07 92 00) at tile; blocking labels ("GRAB BAR BLOCKING 33–36 in") photographed before drywall.

Part 3 — Execution

- **3.01 Examination:** Verify blocking locations from the 06 10 00 photo log; verify tile is cured and finish painting is complete.
- **3.02 Preparation:** Lay out from the fixture centerlines, not the walls; drill tile with diamond bits; apply sealant to penetrations.
- **3.03 Installation:** Mount grab bars 33–36 in AFF to the top of the gripping surface, level, with 1-1/2 in clear to the wall and no obstructions above within 12 in; mount the seat per §608.4; mount mirrors and accessories at the scheduled heights (accessories in reach ranges, never over the grab-bar zone).
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** 250-lb pull test on one bar per cottage; verify heights with a story pole; clean fingerprints; deliver the per-lot blocking record and the grab-bar layout to the HOA for future retrofits.

> ***[REVIEW REQUIRED]** Voluntary program: the Owner shall decide which bars are installed at closing versus blocked-for; confirm current BHMA/ASTM designations.*

Division 12 — Furnishings

Section 12 36 00 — Countertops

Part 1 — General

- **1.01 Section Includes:** Quartz (engineered stone) countertops at kitchens and baths (basis of design), with solid-surface as an approved alternate at baths; undermount sink cutouts; backsplashes (4-in matching or full-height tile per 09 30 13); a 34-in-high accessible work surface option in each kitchen and a 34-in lavatory height at one bath.

- **1.02 Related Sections:** 06 20 00 / 06 41 00 casework (not in this outline); 07 92 00 (sealant at walls and sinks); 09 30 13 (tile backsplash); 22 40 00 (sinks, faucets).
- **1.03 References:** ISFA (International Surface Fabricators Association) Quartz Surfacing Fabrication Standards and Solid Surface Fabrication Standards; NSF/ANSI 51 (food-zone materials); ASTM C880 (flexural strength, engineered stone) and ASTM C1353 (abrasion); GREENGUARD Gold (emissions); ICC A117.1-2017 §1003.12 (Type A kitchen work surface: 30 in wide, 34 in max AFF, knee space) and §606.3 (lavatory rim ≤ 34 in) — voluntary; OSHA 29 CFR 1926.1153 (respirable crystalline silica — quartz fabrication).
- **1.04 Submittals:** 6 × 6-in samples per color; shop drawings with seam locations, sink cutouts, overhangs, and support brackets; edge profile sample; fabricator's silica-control program statement.
- **1.05 Quality Assurance:** Fabricator certified by the slab manufacturer (for warranty) with CNC and water-jet fabrication and wet-cutting only; field-measure after cabinets are set; mock-up: the model cottage kitchen.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Transport on A-frames on edge; store indoors; quartz manufacturer's limited lifetime residential (15-year minimum) warranty; solid surface 10-year; fabricator 2-year (seams, cracks, delamination).

Part 2 — Products

- **2.01 Manufacturers:** Quartz: Cambria (US-made), Caesarstone, Cosentino (Silestone), MSI (Q Premium); solid surface: DuPont/Corian Design (Corian), LX Hausys (HI-MACS), Wilsonart (Solid Surface); or approved equal.
- **2.02 Materials:** Quartz: 3 cm (1-1/4 in) slabs, ≥ 90 percent quartz aggregate in polymer resin, NSF/ANSI 51, GREENGUARD Gold, matte/honed or polished; solid surface: 1/2-in sheet on a built-up edge (1-1/2 in) per ISFA; edges eased (1/8-in radius) or 1-1/4-in square eased — no sharp ogee profiles; backsplash: 4-in matching quartz at baths, tile at kitchens (09 30 13); sinks undermount (22 40 00) with polished cutouts.
- **2.03 Performance Criteria:** Seams ≤ 1/16 in, color-matched epoxy, none within 6 in of a sink cutout; overhangs ≤ 10 in unsupported (steel brackets beyond); kitchen counter height 36 in AFF standard with one 30-in-wide × 34-in section (or a lowered island end) where the plan allows; bath vanity tops at 34 in AFF with a removable-front base cabinet option for knee space; heat resistance per the manufacturer (trivets required — note in the HOA manual); no visible substrate at edges.
- **2.04 Finishes:** Light, low-veining neutrals (illustrative "white with soft gray veining" for kitchens; "warm white" or "light greige" for baths) as selected per scheme; matte finish preferred for glare control in 55+ kitchens.
- **2.05 Accessories:** Sink clips and epoxy; plywood sub-top at dishwasher and sink runs; steel support brackets at overhangs; silicone sealant (07 92 00) at the wall and the sink rim; cutouts for faucets, soap dispensers, and cooktops per the appliance schedule.

Part 3 — Execution

- **3.01 Examination:** Verify cabinets are level within 1/8 in in 10 ft, plumb, and fastened; verify sink and appliance rough-ins.
- **3.02 Preparation:** Template with digital or physical templates after cabinets are set; confirm seam locations with the Architect; shim cabinets, never the top.
- **3.03 Installation:** Set on a continuous bead of silicone on the cabinet rails; level; join seams with color-matched adhesive and seam-setter clamps; mount undermount sinks with clips and silicone; install backsplashes with a 1/8-in sealed joint at the top; cut in place only with wet, HEPA-vacuumed tools.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Straightedge check for flatness and overhang; water-fill test of sink seals; clean with the manufacturer's cleaner; cover with corrugated board until the punch list; provide care instructions to the HOA/homeowner.

  [REVIEW REQUIRED] *Quartz vs. solid surface by room and the accessible-counter option are Owner decisions; confirm the appliance schedule for cutouts.*

Division 22 — Plumbing

Section 22 40 00 — Plumbing Fixtures (fixtures and trim only)

Part 1 — General

- **1.01 Section Includes:** Water closets (comfort height), lavatories and faucets, kitchen sink and faucet, roll-in shower valves and trim with hand-held showers on slide bars, linear drains, optional tub with hand-held shower, laundry box, hose bibbs, water-heater and piping excluded (22 10 00 / 22 30 00 not in this outline).
- **1.02 Related Sections:** 03 30 00 (shower recess, rough-ins); 09 30 13 (tile, drains, waterproofing); 10 28 13 (grab bars, seats); 12 36 00 (sinks in tops); 21 13 00 NFPA 13D (not in this outline).

- **1.03 References:** 2024 International Plumbing Code with Georgia Amendments (2026) — Chapter 4 fixtures, §417 showers (900 in² compartment, 1/4 in/ft floor slope — VERIFY section numbers), §424.3 (shower valves: ASSE 1016 pressure-balance or thermostatic, 120 degrees F maximum), §607 (hot-water temperature); Georgia Water Stewardship Act (O.C.G.A. §8-2-3): high-efficiency fixtures — water closets ≤ 1.28 gpf, lavatory faucets ≤ 1.5 gpm, showerheads ≤ 2.0 gpm (VERIFY current values); EPA WaterSense; ASME A112.19.2/CSA B45.1 (vitreous china), A112.19.3 (stainless sinks), A112.18.1/CSA B125.1 (faucets and showers), ASME A112.6.3 (floor and trench drains); ASSE 1016/A112.18.1 (shower valves), ASSE 1070 (point-of-use temperature limiting at lavatories, optional); ICC A117.1-2017 §604 (water closet height 17–19 in to seat top, centerline 16–18 in from the side wall), §606 (lavatories: rim ≤ 34 in, knee clearance, insulated traps), §608 (roll-in shower controls 38–48 in AFF, hand-held on a 59-in hose with a slide bar), §309.4 (lever/loop handles ≤ 5 lbf, no grasping); NSF/ANSI 372 (lead-free).
- **1.04 Submittals:** Fixture schedule with model numbers, flow rates, WaterSense labels, finishes; rough-in drawings for the linear drain and valve heights; hand-held slide-bar mounting height.
- **1.05 Quality Assurance:** Plumber licensed in Georgia; fixtures from single manufacturers per category for finish matching; the model cottage's primary bath is the mock-up for valve, slide-bar, and drain positions.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store fixtures boxed in the garage until setting; china limited lifetime (defects), faucets/valves limited lifetime (function) and 10-year (finish), drains 5-year; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Kohler; TOTO; American Standard (LIXIL); faucets/showers: Moen, Delta, Kohler; linear drains: Schluter (Kerdi-Line), Infinity Drain, QuickDrain (Oatey); sinks: Elkay, Kohler, Blanco; or approved equal.
- **2.02 Materials:** Water closets: vitreous china, elongated, **comfort/chair height (17–19 in to the top of the seat)**, 1.28 gpf (WaterSense, MaP ≥ 800 g), soft-close slow-close seat, with rear-wall and side-wall grab-bar blocking (06 10 00); lavatories: undermount vitreous china (or integral solid surface at the alternate) with single-lever faucet, 1.2–1.5 gpm, lever handle; kitchen: 18-ga stainless undermount single bowl (30-in) with a pull-down single-lever faucet 1.8 gpm; showers: pressure-balance or thermostatic valve (ASSE 1016) with lever handle set at 38–48 in AFF near the entry, **hand-held shower on a 24-in slide bar (rated as a grab bar where possible) with a 59-in hose and a fixed head via a two-way diverter**, ≤ 2.0 gpm; linear drain 32–48 in stainless with tile-in or brushed grate; optional 60-in acrylic/porcelain-enameled tub (ASME A112.19.1) with a lever-handle valve and hand-held; laundry box with lever ball valves; frost-free hose bibbs (ASSE 1019) with vacuum breakers; insulated P-traps at open-knee lavatories.
- **2.03 Performance Criteria:** All operable parts lever/loop-type, ≤ 5 lbf; faucet handles single-lever; water closet centerline 18 in from the side wall (A117.1 §604.2, 16–18 in) and 60-in turning circle preserved; shower control reachable from outside the spray; hot water ≤ 120 degrees F at showers/tubs (ASSE 1016) and ≤ 110 degrees F at lavatories if ASSE 1070 valves are elected; flow rates per the Georgia Water Stewardship Act and WaterSense.
- **2.04 Finishes:** White china; faucet/shower/accessory finishes per scheme (satin nickel or matte black), matching 10 28 13.
- **2.05 Accessories:** Angle stops (quarter-turn lever), supply lines (braided stainless), escutcheons, wax-free toilet seals, drain assemblies with cleanouts, dishwasher air gap or high loop per the IPC (GA) — VERIFY.

Part 3 — Execution

- **3.01 Examination:** Verify rough-in dimensions (water closet centerline, valve height, drain location and elevation for the 09 30 13 slope), and that tile/waterproofing is complete before trim-out.
- **3.02 Preparation:** Set the linear drain flange level with the membrane per the 09 30 13 system; protect finishes during tile work.
- **3.03 Installation:** Per the IPC (GA) and the manufacturers: water closets on flanges set at the finished floor, caulked at the base except the rear; lavatories with clearances per §606; shower trim with the slide bar fastened into 06 10 00 blocking (not into tile alone); hand-held hose without kinks; aerators installed; hose bibbs with backflow protection.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Verify flow rates and WaterSense labels against the schedule; test the ASSE 1016 limit stop at 120 degrees F; leak test all connections; clean fixtures with non-abrasive cleaner; protect china until turnover.

  [REVIEW REQUIRED] *Life-safety adjacent (scald protection); IPC 2024/GA section numbers, Georgia fixture flow limits, and drain listings are unverified; the plumbing engineer/designer shall complete the schedule.*

Division 26 — Electrical

Section 26 50 00 — Lighting (fixtures only)

Part 1 — General

- **1.01 Section Includes:** Interior LED luminaires (recessed downlights, surface/flush ceiling fixtures, vanity lights, under-cabinet task lights, closet lights, night-lights), porch and garage exterior wall lanterns, driveway/garage-door lights, and the private-lane and pathway post-top lighting; lamps and drivers; controls (dimmers, occupancy/vacancy sensors, photocells, timers). Wiring, panels, and branch circuits are in 26 05 00 / 26 24 00 (not in this outline).
- **1.02 Related Sections:** 07 21 00 (IC-rated fixtures in insulated ceilings); 07 27 00 (air-tight housings); 08 36 13 (opener light); 10 14 00 (sign lighting); 32 16 13 (pole bases); 32 93 00 (light spill on the buffer).
- **1.03 References:** 2023 National Electrical Code with Georgia Amendments (2026) — Article 410 (luminaires), 210.70 (required lighting outlets); 2015 IECC (GA) §R404.1 (≥ 75 percent high-efficacy lamps — VERIFY the GA-amended percentage; specify 100 percent LED); UL 1598 (luminaires), UL 8750 (LED equipment); ENERGY STAR (luminaires) / DLC (exterior); IES RP-28-20 (Lighting and the Visual Environment for Older Adults and the Visually Impaired — design basis for illuminance and glare); IES TM-15 (BUG ratings); IDA-IES Model Lighting Ordinance (MLO); IES RP-8 (roadway/pedestrian lighting — Class P local residential); Lilburn Site Development Plan Review Checklist §5.f (submit a subdivision street-lighting plan and pole/fixture specifications — **VERIFY** the lighting standards that apply outside the Highway 29 Overlay); 2024 IRC (GA 2026) R303 (light and ventilation — interior lighting outlets).
- **1.04 Submittals:** Fixture schedule with photometrics (IES files), CCT/CRI, BUG ratings for all exterior fixtures, lane-lighting point-by-point calculation (average, minimum, max/min, and spill at the property lines/buffer), pole and base details, control schedule; samples of decorative exterior lanterns.
- **1.05 Quality Assurance:** Electrical contractor licensed in Georgia; lane-lighting photometric plan sealed by a Georgia PE or prepared by a lighting-certified designer (LC); night-time aiming and spill check with the Architect before acceptance; decide utility-owned (Georgia Power/Walton EMC lease) vs. HOA-owned lane lighting before design (VERIFY the serving utility).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store fixtures boxed and dry; luminaires 5-year (LED and driver), exterior poles/finish 5-year, decorative lanterns 3-year; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Interior: Lithonia Lighting / Juno / Halo (Acuity — Conyers, GA), Cooper Lighting Solutions (Signify — Peachtree City, GA), WAC Lighting, Progress Lighting, Kichler; exterior lanterns: Hinkley, Kichler, Progress, Troy; lane post-tops: Lithonia (Acuity), Cooper (Signify), Hubbell (Kim/Beacon), Sternberg; or approved equal.
- **2.02 Materials: All lamps and integrated LEDs 3000K CCT (2700K permitted at bedrooms and exterior lanterns), CRI ≥ 90 interior / ≥ 80 exterior**, dimmable (TRIAC/ELV or 0–10 V) with compatible listed dimmers; recessed downlights 4-in or 6-in canless wafer or IC-rated air-tight (ASTM E283 ≤ 2.0 cfm) housings with wet-location trims at showers; surface fixtures with diffusers (no exposed LEDs/bare lamps — glare control per RP-28); vanity: vertical side sconces or a wide diffused bar at 66 in AFF; under-cabinet linear LED at all kitchen counters; closets: surface LED (NEC 410.16 clearances); night-lights: plug-in or switched low-level LED (≤ 1 fc) in the hall and each bath, on a photocell; exterior lanterns: full-cutoff/fully shielded, opaque top, ≤ 3000K, ≤ 800 lm at porches; garage exterior: shielded downlights ≤ 1,000 lm on a photocell/motion; lane: 12–14-ft aluminum poles on concrete bases with full-cutoff post-top or "acorn-style with a full-cutoff optic" LED luminaires, 2,000–3,500 lm, 3000K, BUG ≤ B1-U0-G1, photocell with an astronomic timer and 50 percent dimming after 11 p.m.; pathway/bollard lights at the green and mail kiosk, ≤ 500 lm, fully shielded.
- **2.03 Performance Criteria:** Interior illuminance per IES RP-28 (roughly twice conventional residential levels): kitchen counters 50–75 fc task, bath vanity 30–50 fc at the face (vertical), general living 20–30 fc, hall/stairs (none) 10–20 fc, closets 10 fc; uniformity ≤ 3:1 in circulation; exterior: lane average 0.4–0.6 fc, min 0.1 fc, max/min ≤ 10:1 (RP-8 Class P local, VERIFY Lilburn's standard), **spill ≤ 0.1 fc at the R-1 property lines and buffer, zero uplight (U0)**, all exterior luminaires fully shielded (MLO); porch lights on a switch and a photocell option; occupancy/vacancy sensors at closets, laundry, garage, and baths (fan interlock); rocker/toggle switches at 42–48 in AFF, lighted where useful; all fixtures ENERGY STAR or DLC listed where categories exist.
- **2.04 Finishes:** Exterior lanterns matte black (schemes 1 and 3) or oil-rubbed bronze (scheme 2), as selected; lane poles black; interior trims white or matching hardware.
- **2.05 Accessories:** Compatible dimmers (listed pairings), photocells, astronomic timers, pole bases (32 16 13), in-line fuses at poles, surge protection for lane circuits, spare drivers (5 percent) to the HOA.

Part 3 — Execution

- **3.01 Examination:** Verify ceiling insulation depth for IC/air-tight housings, blocking for lanterns and vanity lights, and pole-base locations clear of sidewalks (≥ 2 ft from the back of curb and out of the sidewalk).
- **3.02 Preparation:** Coordinate fixture locations with the reflected ceiling plans and the sprinkler layout (13D heads); set pole bases with conduit and grounding before paving.
- **3.03 Installation:** Per NEC 410 and the manufacturers: recessed housings sealed to the ceiling gypsum (07 27 00); lanterns on gasketed blocks with the top of the fixture at 66–72 in AFF; poles plumb, luminaires level, photocells facing north; label the

dimmer/driver pairing at each panel; program timers.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Night check: light-meter readings at 10 lane points and at the R-1 property lines against the photometric plan; verify CCT with a meter on a sample of fixtures; verify no uplight; clean lenses and remove labels; deliver O&M data and spare drivers to the HOA.

[REVIEW REQUIRED] Lilburn lighting/light-trespass standards outside the Overlay, the serving utility, and IECC high-efficacy percentage as amended are unverified; the lane photometric plan is a required submittal.

Division 32 — Exterior Improvements

Section 32 12 16 — Asphalt Paving (Private Lane, Guest Bays, and Driveways)

Part 1 — General

- **1.01 Section Includes:** Hot-mix asphalt pavement, graded aggregate base, and subgrade preparation for the 22-ft-wide private lane (in the 31-ft tract; 36-ft tract where widened), hammerhead turnarounds (60-ft legs, 20 ft wide), guest parking bays, the mail-kiosk pull-off, and the Arcado Road entrance apron within the county R/W (per the Gwinnett DOT driveway permit); residential driveways (asphalt or concrete per 32 16 13 at the Owner's option); pavement markings (stop bar, guest-space striping).
- **1.02 Related Sections:** 32 16 13 (curb and gutter, driveway aprons); 26 50 00 (lane lighting); 31 23 00 earthwork and 33 40 00 storm drainage (not in this outline); 32 17 23 pavement markings (included here).
- **1.03 References:** Lilburn Site Development Plan Review Checklist §4.s ("all components of private streets and alleys must meet minimum standards for public street") and §7.i (minimum section for parking areas and drives: **4 in GAB and 2 in Type E or F**), §7.k (street grades ≤ 12 percent; 12–15 percent require an as-graded survey), §7.l (horizontal curves per Gwinnett County standards); **Lilburn Development Regulations (Code Appendix B) — VERIFY the public-street structural section; not retrievable online**; Gwinnett County UDO §900-70.4.A (new local and minor collector streets: ≥ **8 in graded aggregate base, 2 in of 19 mm Superpave intermediate course, 1.25 in of 9.5 mm Superpave Type II surface course**) and §900-70.2.C (fill compacted to 95 percent, top 12 in to 100 percent, of maximum dry density) — used as the presumptive public-street standard; Gwinnett County Standard Drawings (pavement structural sections, Std. 401); GDOT Standard Specifications 2021: Section 310 (GAB), 400 (HMA construction), 412 (tack), 828 (HMA mixes), 815 (GAB material), 820 (asphalt cement PG 64-22 / PG 67-22 as GDOT-specified); ASTM D698 (standard Proctor), D1557 (modified Proctor) — VERIFY which the county uses; ASTM D2950 / D6938 (density); ASTM D6927 (Marshall — not used, Superpave); Lilburn Zoning Ordinance 2023-603 Table 4.2 (pervious paving allowed; front-yard parking discouraged).
- **1.04 Submittals:** Geotechnical pavement recommendations (CBR/subgrade), GAB source certification (GDOT-approved), HMA job-mix formulas (GDOT-approved 19 mm and 9.5 mm Type II), density test reports, plan of the typical section (lane, turnaround, bays, driveways), marking layout.
- **1.05 Quality Assurance:** Paving contractor GDOT-prequalified or with 5 years' subdivision street experience; independent geotechnical testing agency for subgrade, GAB, and HMA density (Division 01 45 00); proof-roll witnessed by the Engineer and the City inspector.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** HMA delivered ≥ 275 degrees F (mix temperature per the JMF), covered trucks; no paving below 50 degrees F ambient (surface course) or on wet base; Contractor 2-year (no rutting > 1/4 in, raveling, cracking, or base failure), matching the City's maintenance-bond period for public streets (VERIFY, since the lane is private and HOA-maintained).

Part 2 — Products

- **2.01 Manufacturers (HMA suppliers):** GDOT-approved plants — e.g., C.W. Matthews Contracting, E.R. Snell Contractor, Pittman Construction / Vulcan Materials; or approved equal.
- **2.02 Materials:** Subgrade: residual soil or engineered fill compacted per §900-70.2.C; GAB: GDOT Section 815 crushed stone; HMA: GDOT Section 828 Superpave 19 mm (intermediate) and 9.5 mm Type II (surface), PG 64-22 or PG 67-22 binder per GDOT, recycled asphalt pavement (RAP) ≤ 25 percent (or per GDOT); tack coat GDOT 412; pavement markings: waterborne traffic paint (GDOT 652) or thermoplastic at the stop bar; pervious concrete/pavers at guest bays as an Owner option (Table 4.2).
- **2.03 Performance Criteria — sections: Private lane, turnarounds, and the entrance (public-street standard, presumptive): 8 in GAB + 2 in 19 mm Superpave + 1.25 in 9.5 mm Type II (3.25 in HMA total)** — VERIFY against the Lilburn Development Regulations at the pre-application meeting; **guest bays and the mail pull-off: 6 in GAB + 2 in 9.5 mm Type II** (exceeds the checklist's 4-in/2-in minimum); driveways (if asphalt): 4 in GAB + 2 in 9.5 mm Type II (checklist §7.i minimum); compaction: subgrade 95/100 percent, GAB 100 percent of ASTM D698 (or D1557 — VERIFY), HMA 92–96

percent of theoretical maximum density (Gmm) per GDOT 400; geometry: 22-ft face-to-face of curb (no parking), design speed 15 mph, 2 percent crown, grades ≤ 12 percent (target ≤ 8 percent), 25-ft minimum centerline radius at turnarounds, hammerheads 60-ft legs × 20 ft; cross-slope 2 percent; ride: ≤ 1/4 in in 10 ft; edge: 24-in curb and gutter (32 16 13); the pavement section shall accommodate a 75,000-lb fire apparatus (IFC 2024 App. D102.1 — VERIFY the Gwinnett Fire requirement for a private lane).

- **2.04 Finishes:** Dense-graded surface; markings: 4-in white guest-space lines, 24-in white stop bar at Arcado Rd, "PRIVATE DRIVE — NO PARKING" signage (Gwinnett Fire) per the site plan.
- **2.05 Accessories:** Geotextile separation fabric (GDOT Section 881, type per the geotechnical engineer) where the geotechnical engineer requires over soft residual clays; underdrains at the swale crossing (u ≈ 700–1,000) if groundwater is encountered; traffic-rated utility frames set flush with the surface course.

Part 3 — Execution

- **3.01 Examination:** Verify all underground utilities (sewer, water, storm, electric, lane-lighting conduit) are installed, tested, and backfilled; verify subgrade elevations and proof-roll acceptance (no pumping under a loaded tandem truck); verify curb and gutter is complete and cured.
- **3.02 Preparation:** Scarify, moisture-condition, and compact the subgrade; remove unsuitable material and replace with GAB or engineered fill per the geotechnical engineer; place GAB in ≤ 6-in lifts to grade ± 1/2 in and prime or keep moist until paving; sweep; apply tack.
- **3.03 Installation:** Per GDOT Section 400 and the JMF: intermediate course 2 in compacted (single lift), surface course 1.25 in compacted, placed with a paver (no hand-spreading except at edges), rolled with breakdown/intermediate/finish passes; joints staggered ≥ 12 in from the intermediate course; longitudinal joints at lane center; match curb-and-gutter flow line within 1/4 in above (never below) the gutter lip; hand-work around drainage structures; markings after 14 days' cure.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Density tests: subgrade and GAB one per 200 ft per lift, HMA cores or nuclear gauge one per 1,000 SY per lift; thickness verified by cores (one per 1,000 SY); straightedge; open to construction traffic only after the intermediate course is placed, and place the surface course after most home construction to avoid construction damage (phased with the two-phase plan); sweep and protect from spills; repair any damage before the surface course.

[REVIEW REQUIRED] *The public-street structural section in the Lilburn Development Regulations could not be retrieved; the Gwinnett UDO §900-70 section is presumptive. The fire-apparatus load requirement for the private lane, the Proctor method, and GDOT section numbers require confirmation by the civil engineer.*

Section 32 16 13 — Concrete Curbs, Gutters, Sidewalks, and Driveways

Part 1 — General

- **1.01 Section Includes:** 24-in concrete curb and gutter (6-in vertical face) along the lane; curb ramps with detectable warnings at the Arcado Road sidewalk and at internal crossings; 5-ft sidewalks (2 ft behind the back of curb; 6 ft where street trees are planted); driveway aprons; concrete driveways and porch/patio walks (where concrete is selected); the mail kiosk pad; pickleball court slab is excluded (32 18 23 by the court contractor — not in this outline); light-pole bases.
- **1.02 Related Sections:** 03 30 00 (house slabs, landings — the walk shall meet the porch landing flush); 26 50 00 (pole bases); 32 12 16; 32 92 00 / 32 93 00 (strip between curb and walk).
- **1.03 References:** Lilburn Site Development Plan Review Checklist (5-ft sidewalks on all roads; 2 ft from the back of curb, 6 ft where street trees; curb ramps and crosswalks at crossings; pole locations); **Lilburn Development Regulations (Appendix B) — VERIFY curb type and sidewalk thickness**; Gwinnett County Standard Drawings (24-in curb and gutter; sidewalk; driveway apron; curb ramp) and UDO §900 — presumptive; GDOT Standard Specifications 2021 Section 441 (concrete curb, gutter, sidewalk, driveways) and Section 500 (Class A concrete, 3,000 psi); GDOT Standard Details 9032B (curb and gutter) and A3 (curb ramps) — VERIFY; ASTM C94, C150, C33, C260 (air entrainment), C309 (curing), D1751 (expansion joint filler), A1064 (WWR where used); U.S. Access Board PROWAG (2023 final rule, 36 CFR 1190) §R304 (curb ramps), §R302 (pedestrian access routes: 2 percent cross slope, 5 percent running or match the roadway grade), §R305 (detectable warning surfaces) — applied by policy on the private lane, mandatory at the Arcado Road public sidewalk; ICC A117.1-2017 §402–403 (accessible routes: ≤ 5 percent running, ≤ 2 percent cross, ≥ 36 in clear, 1/4-in max vertical change) — the design basis for lane-to-porch routes.
- **1.04 Submittals:** Mix design; jointing plan; curb-ramp details with detectable-warning product data; typical sections (curb and gutter, sidewalk, driveway apron); pole-base detail (26 50 00).
- **1.05 Quality Assurance:** Flatwork contractor with 5 years' subdivision experience; ACI-certified flatwork finisher on site; the first 100 ft of curb and 50 ft of sidewalk are the mock-up; slope verification with a smart level at every ramp and landing before acceptance.

- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Concrete placed within 90 minutes of batching; no placement when ambient < 40 degrees F without cold-weather protection; Contractor 2-year (no scaling, spalling, cracking > 1/16 in, or settlement > 1/4 in).

Part 2 — Products

- **2.01 Manufacturers (ready-mix):** Thomas Concrete; Argos USA; Heidelberg Materials; detectable warnings: ADA Solutions, Armor-Tile, Tuftile (cast-in-place composite or cast-iron); or approved equal.
- **2.02 Materials:** Concrete GDOT Class A, 3,000 psi minimum (3,500 psi at driveways and aprons), air-entrained 5–7 percent (ASTM C260; weathering "moderate"), 4-in maximum slump, 3/4-in maximum aggregate; curb and gutter: 24-in-wide, 6-in vertical face, 6-in gutter thickness (Gwinnett/GDOT standard profile — no rolled/mountable curb, which defeats curb ramps and channelization); sidewalk: 5 ft wide × 4 in thick (6 in at driveway crossings), broom finish; driveways/aprons: 6 in thick (4 in at private walks), broom finish, with 1/2-in expansion joints at the curb and the garage slab; light-pole bases: 24-in diameter × 48-in-deep (or per 26 50 00), 3,000 psi, with anchor bolts and conduit; expansion joint filler ASTM D1751 1/2 in; curing compound ASTM C309 Type 1-D; detectable warning surfaces: 24 in deep × full ramp width, contrasting color (federal yellow or brick red vs. gray concrete).
- **2.03 Performance Criteria:** Curb ramps at every crossing: 1:12 (8.33 percent) maximum running slope, 2 percent maximum cross slope, 4-ft × 4-ft landings ≤ 2 percent, flush (0 in) at the gutter, detectable warnings; sidewalks 5 ft minimum clear (no obstructions: poles, hydrants, mailboxes outside the walk), ≤ 2 percent cross slope, running slope matching the lane (≤ 5 percent preferred; the lane's ≤ 12 percent grade limit governs and steeper segments are permitted where they match the road grade per PROWAG); **walks from the lane sidewalk to each porch: ≤ 5 percent running, ≤ 2 percent cross, ≥ 36 in (48 in preferred) wide, meeting the porch landing with ≤ 1/4-in vertical change (zero-step route);** driveway aprons with the sidewalk carried across at ≤ 2 percent cross slope (the apron slope occurs between the curb and the walk); joints: sidewalk contraction joints at 5 ft, expansion at 50 ft and at structures; curb contraction joints at 10 ft, expansion at 50 ft and at radii returns; curb ramps and pole bases doweled to the curb.
- **2.04 Finishes:** Medium broom perpendicular to travel; tooled edges 1/4-in radius; curb face steel-troweled then lightly broomed; decorative (exposed aggregate/colored) walks at the village green only with the Director's approval (checklist §5.b analog — VERIFY).
- **2.05 Accessories:** Weep pipes through the curb for lot drainage where the grading plan requires; steel plate at storm inlets (curb inlet throats by 33 40 00); 2-ft × 8-ft amenity pads every 300 ft along the Arcado Road frontage if required (checklist §5.b is written for the Overlay — VERIFY applicability); mail-kiosk pad 6 in thick with a 5-ft clear approach.

Part 3 — Execution

- **3.01 Examination:** Verify subgrade compaction (95 percent) and 4-in GAB or compacted stone base under curb and gutter (2 in of stone under walks where the geotechnical engineer requires); verify forms/slipform stringline to the design flow line ± 1/4 in; verify ramp and landing layout against the accessible-route plan.
- **3.02 Preparation:** Set forms or slipform; place expansion joints and dowels; wet the base; install detectable-warning setting frames; stake pole bases from the 26 50 00 layout.
- **3.03 Installation:** Place, consolidate, screed, and float; broom finish; tool joints within 12 hours or saw within 24; cure 7 days with compound or wet cover; strip forms after 24 hours and backfill within 7 days; set detectable warnings in fresh concrete; slope every ramp and landing with a 4-ft smart level before the concrete sets and correct while plastic.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Cylinders per 50 cy; slope audit of every ramp, landing, and porch approach recorded on a checklist (a slope > 2.0 percent cross or > 8.33 percent ramp is rejected and replaced); flow-line water test after the surface course; keep construction traffic off walks; protect curb from asphalt tack overspray; repair scaling by replacement, not patching.

  [REVIEW REQUIRED] *Curb profile, sidewalk thickness, and ramp details are presumptive Gwinnett/GDOT standards pending the Lilburn Development Regulations; the accessible-route slopes on the lots with cross-slope (NE to SW) need the civil engineer's grading plan.*

Section 32 31 13 — Chain Link Fences and Gates (Pickleball Court Enclosures)

Part 1 — General

- **1.01 Section Includes:** Black vinyl-coated chain-link fencing, 10 ft high, enclosing each 30-ft × 60-ft pickleball pad (two courts, each a 20-ft × 44-ft playing area), with accessible pedestrian gates, a maintenance gate, and optional sound-attenuating and wind-screen panels; fence posts and footings; court surfacing, nets, and posts are by 32 18 23 (not in this outline).
- **1.02 Related Sections:** 03 30 00 / 32 16 13 (court slab and approach walk — by the court contractor); 26 50 00 (no court lighting is proposed — daylight play only; VERIFY); 32 93 00 (screen planting between the courts and adjacent R-1 lots); 10 14 00 (rules signage).

- **1.03 References:** Lilburn Zoning Ordinance 2023-603 — accessory structure and fence height/setback provisions (**VERIFY: a 10-ft sport-court fence may exceed the residential fence height limit and require a specific approval or condition; keep the courts out of the 20-ft buffer and ≥ 5 ft from it — checklist "≥ 5-ft parking setback from buffers" analog**); ASTM F668 (PVC-coated chain-link fabric, Class 2b fused-and-bonded), ASTM F1043 (framework strength, Group IC) and ASTM F1083 (Schedule 40 pipe), ASTM F626 (fittings), ASTM F900 (gates), ASTM F567 (installation practice), ASTM F1183 (aluminum alternate — not used), ASTM A123/A153 (galvanizing); USA Pickleball Official Rulebook §2 (court dimensions: 20 × 44 ft, recommended minimum 30 × 60 ft playing surface, 34-in net at center) and USA Pickleball / ASBA Pickleball Courts: Construction & Maintenance Manual (fence heights, windscreens, acoustic fencing guidance); ICC A117.1-2017 §404 (gates on accessible routes: 32-in clear, lever/loop hardware, ≤ 5 lbf, no threshold).
- **1.04 Submittals:** Fence layout with post spacing and gate locations; product data for fabric, framework, fittings, gates, hardware, and acoustic/windscreen panels (with wind-load data); footing detail; color sample.
- **1.05 Quality Assurance:** Fence contractor with 5 years' sports-fence experience; the first court's fence reviewed before the second; note that pickleball noise (≈ 70 dBA at 100 ft) is the most common 55+/neighbor complaint — the amenity block location, orientation, acoustic panels, and HOA hours (e.g., 8 a.m.–8 p.m.) are the mitigation, offered as a voluntary zoning condition.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store fabric rolls on end, framework off the ground; PVC coating 15-year (chalk/peel), framework 10-year, gates and hardware 5-year, acoustic panels 10-year; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Chain link systems: Master Halco, Merchants Metals, Ameristar (ASSA ABLOY) — ornamental alternate at the entry/green; acoustic panels: Acoustifence (Acoustiblok), Sound Fighter Systems, Kinetics Noise Control (VERIFY sports-fence products); windscreens: Aer-Flo, Putterman; or approved equal.
- **2.02 Materials:** Fabric: 9-ga core (0.148 in) with 6-ga finished PVC-coated (Class 2b fused and bonded) black, 1-3/4-in mesh (ball containment; 2-in acceptable), knuckled top and bottom selvage (no barbs), 10-ft height at the end (baseline) fences and full 10 ft at the sides (4-ft side fences optional at the Architect's choice — full height specified for containment and screen mounting); framework ASTM F1043 Group IC (or F1083 Sch. 40), galvanized and black PVC/polyester coated: terminal/corner/gate posts 3-in OD (10-ft fence, set 42 in deep), line posts 2-1/2-in OD at ≤ 10 ft o.c., top and bottom rails 1-5/8-in OD, mid rail at 5 ft; fittings F626 black; tension wire 7-ga at the bottom rail if no bottom rail; gates F900: two 4-ft-wide single swing pedestrian gates per pad (≥ 32 in clear, self-closing with a lever/loop latch operable ≤ 5 lbf, no drop rod at the walking surface, 1/2-in maximum threshold — ideally 0), one 6-ft double maintenance gate per pad; hardware black; optional windscreens (open-mesh, ≤ 70 percent blockage, with wind-load relief) and optional mass-loaded vinyl acoustic panels (≥ 1 lb/SF, STC ≥ 28) on the sides facing the nearest R-1 lots — post spacing and footings sized for the panel wind load if installed.
- **2.03 Performance Criteria:** Wind: framework and footings designed for 115 mph Exposure B (VERIFY) with screens/panels attached (typical: 12-in-dia × 42-in footings at line posts, 16-in × 48-in at terminals — Engineer to confirm); fabric taut without sag (≤ 1 in deflection under a 50-lb pull at mid-span); bottom of fabric ≤ 2 in above the court surface; gate clear opening 32–36 in; no sharp edges (knuckled selvage; bolt ends peened); court pad drains at 1 percent minimum away from the fence (by 32 18 23); fence located ≥ 5 ft from the buffer and outside all setbacks (VERIFY per the Zoning Ordinance).
- **2.04 Finishes:** All components black (PVC-coated fabric; polyester-powder or PVC-coated framework and fittings); hardware black.
- **2.05 Accessories:** Post caps, rail ends, tension bars and bands, truss rods at gate posts; rules/hours sign (10 14 00) at each gate; bench pads and a shade structure by others; ball-containment "kick plate" (1-ft-high solid panel) at the bottom of the end fences optional.

Part 3 — Execution

- **3.01 Examination:** Verify the court layout, pad edges, and gate locations against the amenity plan and the accessible route from the clubhouse/mail kiosk; verify setbacks and the buffer line from the survey.
- **3.02 Preparation:** Call 811; auger footings; set posts plumb in 3,000-psi concrete (crowned at grade); cure 3 days before stretching.
- **3.03 Installation:** Per ASTM F567: rails and mid-rail, then fabric stretched with a come-along and tied at 12 in o.c. on rails and 24 in on posts (aluminum ties, black), knuckled ends; gates hung plumb with hinges adjusted for self-closing and latch alignment; screens/panels with grommets at 12 in o.c. and tie-wraps rated for UV, leaving relief gaps per the manufacturer.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Plumb (± 1/4 in in 10 ft) and height checks; gate operation and latch force test; touch up coating scratches with the manufacturer's paint; remove spoil and concrete splatter from the court pad before surfacing.

> ***[REVIEW REQUIRED]*** *Fence height versus the Lilburn residential fence limit, wind design with screens, and the noise-mitigation package are unverified and should be raised at the pre-application conference.*

Section 32 92 00 — Turf and Grasses (Sod)

Part 1 — General

- **1.01 Section Includes:** Topsoil placement and fine grading; warm-season sod on all lots (front, side, and rear yards outside the buffer and preserved woods), the village green, lane strips, and pond embankments above the normal pool; temporary and permanent seeding of disturbed areas outside sod limits (by 31 25 00 erosion control — coordinate); initial establishment; irrigation by 32 84 00 (not in this outline).
- **1.02 Related Sections:** 32 16 13 (walk edges); 32 93 00 (beds and buffer supplement); 31 25 00 (erosion and sediment control — Georgia "Green Book"), 33 40 00 (pond grading) — not in this outline.
- **1.03 References:** Georgia Crop Improvement Association (GCIA) Certified Sod program; TPI (Turfgrass Producers International) Guideline Specifications to Turfgrass Sodding; UGA Cooperative Extension turfgrass recommendations for the Georgia Piedmont; ASTM D5268 (topsoil for landscaping); Manual for Erosion and Sediment Control in Georgia (current ed.) for temporary cover; Lilburn Site Development Plan Review Checklist (landscape strip and buffer planting; tree protection); Gwinnett County outdoor water-use rules (irrigation schedules — VERIFY).
- **1.04 Submittals:** Sod source and GCIA certificate per variety; topsoil analysis (UGA soil test: pH, organic matter, texture) and amendments; sod layout by variety; establishment schedule; the HOA maintenance plan (mowing height, fertilization, irrigation).
- **1.05 Quality Assurance:** Landscape contractor licensed in Georgia with a GCLP (Georgia Certified Landscape Professional) on staff; sod cut ≤ 24 hours before laying (36 in cool weather); mock-up: the model-cottage lots.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver sod on pallets, keep shaded and moist, do not stack > 24 hours; 90-day establishment period with replacement of any area not 95 percent covered and rooted; Contractor 1-year on grading and drainage (no ponding > 24 hours).

Part 2 — Products

- **2.01 Sources:** GCIA-certified Georgia sod farms (e.g., Super-Sod / Patten Seed, NG Turf, Legacy Turf Farms — Georgia growers); or approved equal.
- **2.02 Materials:** Full-sun lots, green, and lane strips: 'TifTuf' hybrid bermudagrass (drought-tolerant, GCIA certified) or 'Zeon' / 'Emerald' zoysiagrass (finer texture, shade-tolerant to ~4 hours sun — preferred at front yards for a cottage character); shaded rear yards under retained trees and buffer edges: turf-type tall fescue blend (Georgia-recommended cultivars) sod, or convert to mulched native groundcover beds (32 93 00) where sun < 4 hours; topsoil: ASTM D5268, sandy loam to loam, pH 6.0–6.5, ≥ 3 percent organic matter, 4 in placed at all sodded areas (6 in at the green); starter fertilizer per the soil test; lime per the soil test (Piedmont clays are typically acidic).
- **2.03 Performance Criteria:** Finish grade 1 in below walks and curbs (sod thickness), sloping ≥ 2 percent away from foundations for 10 ft (IRC R401.3: 6 in in 10 ft) — **at zero-step entries, achieve fall with swales alongside the landing rather than by raising the landing**; no slopes > 3:1 in sodded areas (use 32 93 00 plantings or fescue with erosion netting on steeper pond banks); sod strips staggered, butted, no gaps > 1/4 in, rolled, watered within 30 minutes; establishment: 95 percent cover and rooted (cannot be lifted) at 90 days; mowing height per variety (bermuda 1–1.5 in, zoysia 1.5–2 in, fescue 3–3.5 in).
- **2.04 Finishes:** Uniform variety per street frontage (no mixed varieties on adjacent lots); mulch (32 93 00) at all bed edges with a spade-cut edge.
- **2.05 Accessories:** Sod staples on slopes > 4:1; erosion-control blanket (Georgia Green Book Type) under fescue on pond banks; temporary irrigation where the permanent system is not yet live.

Part 3 — Execution

- **3.01 Examination:** Verify rough grades ± 0.1 ft, positive drainage, utility trenches settled, and the buffer/tree-protection fencing in place; verify the soil test results are in hand.
- **3.02 Preparation:** Rip compacted subsoil 6 in (except within tree drip lines); spread and rake topsoil; incorporate lime/fertilizer; fine grade; water to settle; remove stones > 1 in and debris.
- **3.03 Installation:** Lay sod on moist soil within 24 hours of cutting, perpendicular to slopes, staggered joints, tight seams, rolled, and watered to 4-in depth; keep off for 2 weeks; first mowing when rooted; install warm-season sod March–October (dormant sod November–February accepted only with the Architect's approval and extended establishment); fescue September–November or February–April.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect at 30, 60, and 90 days for cover, weeds, and drainage; replace failed areas; clean walks and curbs of soil daily; protect with temporary fencing at the model lots until turnover to the HOA with the maintenance plan.

> ***[REVIEW REQUIRED]*** *Variety selection depends on the lot-by-lot sun exposure after clearing; irrigation and water-use rules are unverified; the erosion-control sequencing is by the civil engineer's ES&PC plan.*

Section 32 93 00 — Plants (Buffer Supplement, Landscape Strip, Street Trees, and Foundation Plantings)

Part 1 — General

- **1.01 Section Includes:** Supplemental evergreen and canopy planting within the 20-ft perimeter buffer where the existing vegetation is sparse (no clearing or grading in the buffer); the 10-ft landscape strip along Arcado Road; lane street trees; screen planting at the pickleball courts, ponds, and mail kiosk; foundation and porch plantings at each cottage (HOA-maintained); native/adapted Georgia Piedmont species; mulch, staking, and a 1-year maintenance/warranty period. Irrigation by 32 84 00; tree protection by 01 56 39 / 31 10 00 (not in this outline).
- **1.02 Related Sections:** 10 14 00 (sign bed); 26 50 00 (light spill and pole/tree conflicts); 32 16 13 (root barriers at walks); 32 31 13 (court screen); 32 92 00 (topsoil, sod edges).
- **1.03 References:** Lilburn Zoning Ordinance 2023-603 §313 (buffers: undisturbed, supplemented to an effective screen — **VERIFY the planting standard and the "opaque within 3 years" language**) and Table 4.1 (20-ft buffer abutting R-1 for cottage homes); Lilburn Site Development Plan Review Checklist: 10-ft landscape strip along all public R/W with planting at the stated rate ("for each 10-foot LS strip install 1 tree and 1 shrub" per unit length — VERIFY the rate), tree protection for trees > 5-in caliper in buffers and strips, street-tree list (Willow Oak, Overcup Oak, Nuttall Oak, Pin Oak, Shumard Oak, Lacebark Elm, Japanese Zelkova — written for the Overlay; used as the accepted list), 4-in caliper at 6 ft from the back of curb for street trees along the public R/W; ANSI Z60.1-2014 (American Standard for Nursery Stock); ANSI A300 Part 6 (planting and transplanting); ISA BMPs; Georgia Exotic Pest Plant Council (GA-EPPC) Category 1 invasive list (prohibited); UGA Extension "Native Plants for Georgia" (Parts I–IV) — the species basis; Georgia Department of Agriculture nursery certification.
- **1.04 Submittals:** Planting plan with a schedule (botanical/common name, size, quantity, root condition, spacing); nursery certificates and source list (Georgia-grown preferred); soil test and amendments; mulch sample; the buffer inventory (existing trees > 5 in caliper, canopy gaps) that drives the supplement quantities; the HOA maintenance plan.
- **1.05 Quality Assurance:** Landscape contractor with a GCLP; plants inspected at the nursery or on delivery for Z60.1 conformance; ISA-certified arborist to mark buffer gaps and supervise work within tree drip lines; the model-cottage frontage is the foundation-planting mock-up.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver plants within 48 hours of digging (B&B) or in containers; protect from wind and sun; keep root balls moist; plant within 3 days of delivery; 1-year warranty with one replacement of dead or defective plants at the end of the period; 90-day establishment maintenance (watering, weeding, resetting) included.

Part 2 — Products

- **2.01 Sources:** Georgia Green Industry Association member nurseries (e.g., Bold Spring Nursery, Hackney Nursery, Select Trees, Saunders Brothers — regional growers); or approved equal.
- **2.02 Materials — species palette (Georgia Piedmont natives and proven adapted cultivars; final selections by the Landscape Architect):**
- Canopy (buffer supplement and green): *Quercus alba* (white oak), *Q. phellos* (willow oak), *Q. shumardii* (Shumard oak), *Q. lyrata* (overcup oak), *Nyssa sylvatica* (black gum), *Liriodendron tulipifera* (tulip poplar, buffer only), *Acer rubrum* (red maple) — 2-1/2–3-in caliper B&B in the buffer; 4-in caliper at the Arcado Road strip per the checklist.
- Evergreen screen (buffer gaps, courts, ponds, kiosk): *Ilex opaca* (American holly), *Juniperus virginiana* (eastern red cedar, sunny edges), *Magnolia virginiana* (sweetbay), *Magnolia grandiflora* 'Little Gem' (adapted cultivar), *Ilex vomitoria* (yaupon, adapted), *Osmanthus americanus* (devilwood) — 6–8 ft height B&B, double staggered rows at 8–10 ft o.c. where the buffer is open.
- Understory/ornamental: *Cercis canadensis* (redbud), *Cornus florida* (dogwood, part shade), *Amelanchier arborea* (serviceberry), *Chionanthus virginicus* (fringetree), *Oxydendrum arboreum* (sourwood), *Hamamelis virginiana* (witch hazel) — 1-1/2–2-in caliper or 6–8 ft.
- Shrubs: *Hydrangea quercifolia* (oakleaf hydrangea), *Itea virginica* 'Henry's Garnet', *Ilex glabra* (inkberry, 'Shamrock'), *Callicarpa americana* (beautyberry), *Fothergilla gardenii*, *Viburnum dentatum / V. nudum*, *Clethra alnifolia*, *Rhododendron canescens* (Piedmont azalea, shade), *Ilex verticillata* (winterberry, wet edges) — 3- and 7-gal.
- Groundcovers/perennials: *Carex pensylvanica*, *Chasmanthium latifolium* (shade), *Muhlenbergia capillaris* (muhly grass), *Rudbeckia fulgida*, *Echinacea purpurea*, *Amsonia hubrichtii*, *Packera aurea*, *Phlox divaricata* — 1-gal and plugs; pond banks above the pool: native seed mix (Georgia Green Book) with *Juncus*, *Carex*, *Panicum virgatum*.
- Lane street trees (in the 5-ft lawn strip only where the 36-ft tract allows a ≥ 6-ft strip; otherwise in lot front yards): *Q. phellos*, *Q. shumardii*, *Ulmus parvifolia* 'Allee' (lacebark elm), *Zelkova serrata* — 3-in caliper, 40 ft o.c. average, ≥ 6 ft from the back of curb,

≥ 10 ft from lane lights and ≥ 15 ft from sewer laterals.

- **Prohibited:** GA-EPPC Category 1 species — *Ligustrum* spp., *Nandina domestica*, *Pyrus calleryana* (Bradford pear), *Elaeagnus* spp., *Lonicera japonica*, *Hedera helix*, *Wisteria sinensis*, *Vinca* spp.; *Lagerstroemia* (crape myrtle) is adapted and permitted outside the buffer.
- **2.03 Performance Criteria:** Buffer supplement quantity: planted only in gaps mapped by the arborist to achieve a continuous 6-ft-high visual screen within 3 growing seasons (≈ 1 evergreen per 8–10 lineal ft of gap plus 1 canopy tree per 30–40 ft — VERIFY against §313); landscape strip: 1 canopy tree (4-in cal.) and shrubs per the checklist rate per 10 ft (VERIFY); foundation plantings: low (≤ 30 in mature) shrubs under windows, no thorny or high-pollen species at porches, ≥ 3 ft clear of the siding for the 6-in siding-to-grade clearance and HVAC access, ≥ 18 in clear of walks; sight triangles at the Arcado Road entrance kept ≤ 30 in high (VERIFY the Gwinnett DOT triangle); all plants Z60.1 with a well-developed root system, no circling roots, single leader on canopy trees; planting soil per the soil test; mulch 3 in (pine straw or shredded hardwood) held 3 in from trunks; no mulch volcanoes.
- **2.04 Finishes:** Bed lines spade-cut; steel edging optional at the green; tree grates not used; root barriers (18-in deep) at street trees within 5 ft of walks.
- **2.05 Accessories:** Staking only where needed (remove at 1 year); trunk guards (deer/mower); tree-watering bags for the first season; slow-release fertilizer tablets; rodent/deer repellent as needed; tree-protection fencing (orange, at the drip line) for retained trees and the buffer during construction (checklist requirement).

Part 3 — Execution

- **3.01 Examination:** Verify the buffer inventory and gap map, tree-protection fencing, finish grades (32 92 00), irrigation stubs, and utilities (811); verify the lane lights, laterals, and sight triangles are staked.
- **3.02 Preparation:** Lay out and obtain the Landscape Architect's approval of locations; dig holes 2–3 times the root-ball width and no deeper than the ball (root flare at or 1 in above grade); no digging within the drip line of retained buffer trees except by hand for the supplement plants; amend backfill only per the soil test (no wholesale replacement).
- **3.03 Installation:** Per ANSI A300 Part 6: remove wire baskets (top 2/3) and burlap; set plumb; backfill in lifts with water; form a 3-in saucer; mulch; stake only in windy exposures; prune only dead/broken branches; water thoroughly and weekly (1 in) through the first season; plant trees October–March preferred (containers year-round with irrigation).
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect at planting, 30 days, and the end of the 1-year period; replace failures at the next planting season; remove tags and stakes at 1 year; keep beds weed-free through establishment; the HOA's landscape contract takes over at turnover with the plan and warranty documentation.

  [REVIEW REQUIRED] *Buffer supplement rate, landscape-strip rate, street-tree placement in the 31-ft tract, and species suitability per the post-clearing sun/soil conditions require the Landscape Architect and the Lilburn buffer standard (§313 / Development Regulations).*

Summary and review flags

Specifications written: 30 sections across 11 divisions. Sections flagged **[REVIEW REQUIRED]**: 30 of 30 (every section is outline-level with generic manufacturers; the reasons cluster as follows).

1. **Code edition change (all sections):** Georgia's mandatory codes changed on 1 January 2026 to the **2024 IRC / IBC / IPC / IMC / IFGC with 2026 Georgia Amendments, the 2023 NEC (2026 GA Amendments), and the 2024 IFC** — not the 2018 editions assumed in the task brief and in `FACTS.md` §2 ("IFC 2018 App. D"). Section numbers cited follow the 2018–2024 numbering and must be re-verified against the 2024 texts. The **energy code remains the 2015 IECC with Georgia Amendments (2023)**: Zone 3 wall R-13 (not R-20/R-13+5), ceiling R-38 (R-30 alternative), fenestration U-0.35/SHGC-0.27, ≤ 5 ACH50 — the design (R-20 walls, U ≤ 0.28 / SHGC ≤ 0.23 windows, ≤ 3 ACH50 target) exceeds these minimums.
2. **Life-safety-adjacent sections (Architect/AHJ confirmation):** 08 14 00 (egress door, garage separation door — self-closing/self-latching, safety glazing, zero-threshold water management vs. IRC R311.3), 08 36 13 (garage-door wind-load label, UL 325 opener), 08 53 13 (bedroom egress net-clear openings, safety glazing, PG rating), 09 29 00 (garage/house separation, fireblocking, ceiling board on 24-in trusses), 07 31 00 (Class A, wind attachment), 22 40 00 (ASSE 1016 scald protection), 03 30 00 / 06 10 00 (all structural values pending the geotechnical report and a Georgia PE).
3. **Local standards not retrievable — VERIFY at the Lilburn pre-application conference:** the **Lilburn Development Regulations (Appendix B) public-street structural section** (32 12 16 uses Gwinnett UDO §900-70: 8 in GAB + 2 in 19 mm + 1.25 in 9.5 mm Superpave as presumptive; the checklist's 4-in GAB / 2-in Type E-F minimum applies to drives and parking), curb profile and sidewalk thickness (32 16 13), the §313 buffer-supplement planting standard and the landscape-strip planting rate (32 93 00), the sign article's area/height/setback/illumination limits for a ≤ 32-SF monument sign and the separate sign permit (10 14 00), the residential fence-height limit vs. a 10-ft sport-court fence and the pickleball noise package (32 31 13),

lighting/light-trespass standards outside the Highway 29 Overlay and the serving utility (26 50 00), and Gwinnett Fire's apparatus-load requirement for the private lane (32 12 16).

4. **Zero-step entry detail set (03 30 00, 04 40 00, 07 46 46, 08 14 00, 32 16 13, 32 92 00):** the 1/2-in maximum rise at every door is reconciled with IRC wood/siding clearances by stopping siding on the masonry water table (stone may be 1/2 in above a walking surface on the same foundation, IRC R703.12.1), sloping landings 2 percent, draining with swales rather than raised landings, and specifying ADA-type sills under a 6-ft porch roof; the Architect must draw these five details before the elevations are finalized.
5. **Owner selections still open:** factory ColorPlus vs. field paint (07 46 46 / 09 91 13); manufactured vs. natural stone (04 40 00); SPC vs. WPC plank (09 65 00); quartz vs. solid surface by room (12 36 00); which grab bars are installed at closing vs. blocked-for (10 28 13); utility-leased vs. HOA-owned lane lighting (26 50 00); sod variety by lot exposure (32 92 00); ENERGY STAR climate zone for Gwinnett County (08 53 13). All colors and hex values in this document are illustrative placeholders.

Sources

- Georgia DCA, "Current State Minimum Codes for Construction" — https://dca.georgia.gov/community-assistance/construction-codes/current-state-minimum-codes-construction (accessed 2026-08-28: 2024 IRC/IBC/IPC/IMC/IFGC/ISPSC with 2026 GA Amendments; 2023 NEC; 2024 IFC; 2015 IECC with GA Supplements and Amendments, 2023); DCA announcement "New Georgia Codes and Amendments — Effective January 1, 2026" — https://dca.georgia.gov/announcement/2025-12-09/new-codes-jan-2026
- 2015 IECC with Georgia Amendments, Zone 3 values — Insulation Institute summary https://insulationinstitute.org/wp-content/uploads/2023/02/GA_Code_2023_v2.pdf and UpCodes, Georgia State Minimum Standard Energy Code 2015, Chapter RE 4 — https://up.codes/viewer/georgia/iecc-2015/chapter/RE_4/re-residential-energy-efficiency
- Gwinnett County UDO §900-70 Street Construction Standards — http://gwinnettcounty.elaws.us/code/coor_apxa_ch900_sec900-70; City of Lilburn Site Development Plan Review Checklist — https://www.cityoflilburn.com/DocumentCenter/View/1888 (local copy `data/site-dev-plan-review-checklist.txt`)
- Lilburn Zoning Ordinance 2023-603 — https://www.cityoflilburn.com/DocumentCenter/View/3664 (excerpts in `data/ordinance-excerpts.md`); project facts: `status/cottage-submission-2026-08-28/FACTS.md` §2, §3, §4, §6 (2026-08-28)
- Product and industry standards cited by designation in each section (ASTM, ANSI, TCNA, SMACNA, MPI, DASMA, NAFS, ICC A117.1-2017, IES RP-28, ASTM F3261, GDOT Standard Specifications 2021) — editions to be confirmed by the specifier of record.

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